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Behavioral Games and Water-Saving Technology Adoption in Uzbekistan

Anton Liutin* Jamshid Shukurullaev†

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Abstract

Uzbekistan subsidizes drip irrigation to save water. We study who adopts drip along a shared canal, and whether demand arises where adoption does the most for the group. In a framed field experiment, 190 farmers in 38 five-person groups share a simulated canal. Water flows from head to tail, and farmers decide in turn in full view of the group. We print the payoff cards and reassign seats at random between sessions. Randomized position doubles demand: 29 percent of head farmers buy drip at the first decision, against 58 percent at the tail. Buying is a bet on how long play lasts and, at the tail, on upstream farmers collectively leaving enough water. Because the game gives everyone the same payoffs, rules and price, a common private benchmark describes common strategy. It reproduces demand by seat but leaves most of the variation in who buys unexplained. Of the variance in individual purchases, 41 percent persists within farmers after seats are reassigned, and 16 percent is shared within groups. Private and group incentives to adopt pull apart where water is plentiful. Upstream, the buyer's own gain from drip is smaller than buying requires. With plentiful water, however, drip makes sharing cheap, and in a calibrated canal model the group production gain of an upstream installation is larger, with a third of it going to the farmers below. Demand instead concentrates at the tail, where under typical sharing the group production gain is smallest. Groups with more upstream adopters deliver more water downstream, but the farmers below absorb almost all of a release, so total extraction falls only slightly. Farmers who report using drip on their own farms took less water in the game's practice session. Farmers can buy efficiency, but not the allocation that decides whether it helps them.

1 Introduction

Economists usually explain who adopts a farm technology by its return to the farmer who buys it, and by how farmers learn that return from experience and from

*University of Wisconsin-Madison, Department of Agricultural and Applied Economics. Email: liutin@wisc.edu
†Bukhara State Technical University

neighbors [12, 9, 28]. Most technologies affect only the farm that installs them. For some technologies, however, one farmer's installation directly affects other farms too. Drip irrigation on a shared canal is an example. A farmer who installs drip needs less water for the same crop, and the water saved can go to other farmers [13]. What drip is worth to its buyer and what it does for others therefore need not match. We ask whether the farmers who want drip are the ones whose adoption helps others.

Governments across water-scarce regions promote drip irrigation to save water. Uzbekistan made it a central response to scarcity. Renewable water per person fell by a quarter in the last 20 years, and agriculture takes more than 90 percent of the water withdrawn [33]. The government promotes drip to keep farming going with less water and to cut the electricity used to pump water to the fields [17]. State subsidies of 8–12 million Uzbek som per hectare covered 30–40 percent of installation costs [7]. In 2026 alone, the government set aside 800 billion som in subsidies and 2.6 trillion som in bank loans for water-saving technologies, together about $280 million [26]. These subsidies, however, rely on voluntary take-up. The policy counts on the farmers who install drip to take less water, so that the water they save can keep other farms going. Studies of the irrigation efficiency paradox find that farmers often use the saved water to irrigate more land, so total water use rarely falls [13, 25].

A second reason holds even when no land is added. It lies in who wants drip. Along a shared canal, farmers do not control how much water reaches them. Their supply depends on what farmers upstream take [23, 24].[1] A drip system makes the water a farmer receives more productive, but it cannot secure the water itself. That shapes who wants the technology. Farmers near the head already have plenty of water, so drip adds little to their own harvest. But their adoption is what would release water for the farmers below. Farmers near the tail are short of water, so they want drip to stretch what little they receive. Their adoption, however, frees no water for anyone else. The farmers who ask for drip are therefore not the ones whose adoption helps the rest of the canal.

[1] In our 2023 focus-group discussions with 19 farmers who had not installed water-saving technologies, insufficient water and unreliable electricity were among the reported barriers.

We test the argument in a framed field experiment in Uzbekistan. It holds when water is plentiful. Farmers at the tail of the canal buy drip most often, although their adoption helps no one else. Farmers upstream buy it least often, although their adoption would help others most. We adapt the irrigation game [15, 6] to add a decision to buy drip. We play it with 190 farmers from four regions, recruited from a survey of 914 farms. Farmers play in 38 groups of five along a simulated canal. Water flows from the head (seat 1) to the tail (seat 5), and each farmer can use only what upstream farmers leave. Farmers decide in turn from head to tail, and everyone in the group sees every decision. We reassign seats at random between sessions and record every withdrawal.

The game also lets us separate why farmers buy drip irrigation. A farmer may buy because of the seat, because of who the farmer is, or because of the group. In the field, these reasons are hard to tell apart, because the return to drip also differs with soil, water, markets and skill. In the game, every farmer gets the same payoff cards, rules and price. What differs is the seat, which decides how much water a farmer can expect. We can therefore compute what any farmer who wants to earn the most would do in each seat. We call this the private benchmark. What the benchmark does not explain comes from the farmer or from the group. Because seats are reshuffled between sessions while groups stay together, we can tell the two apart.

A random tail seat doubles demand for drip. When drip first becomes available, 11 of 38 farmers at the head buy it (29 percent), against 22 of 38 at the tail (58 percent). Seats are assigned at random, so the seat causes the gap. Demand rises by about 7 percentage points with each seat toward the tail (exact randomization test, $p = 0.002$).

Farmers upstream buy least because drip adds little to their own pay. With plentiful water, drip adds about 5 tokens a round near the head, just short of what it needs to pay back. Yet these are the farmers whose adoption helps the group most. On drip, a farmer near the head loses little by taking only a fair share, and the water left flows to the farmers below. We calibrate a model of the canal on the games and use it to measure the group production gain, what an installation adds to the output of all five farmers. An installation at seats 1–3 adds about 8 tokens

a round, and a third of that goes to the farmers below. An installation at the tail adds about 3 tokens, all of them its buyer's own. Demand is highest where this gain is smallest and lowest where it is largest (Figure 1). The gap is a feature of plentiful water. When supply falls in the next session, the strongest incentive moves up to seat 4, which then buys the most.

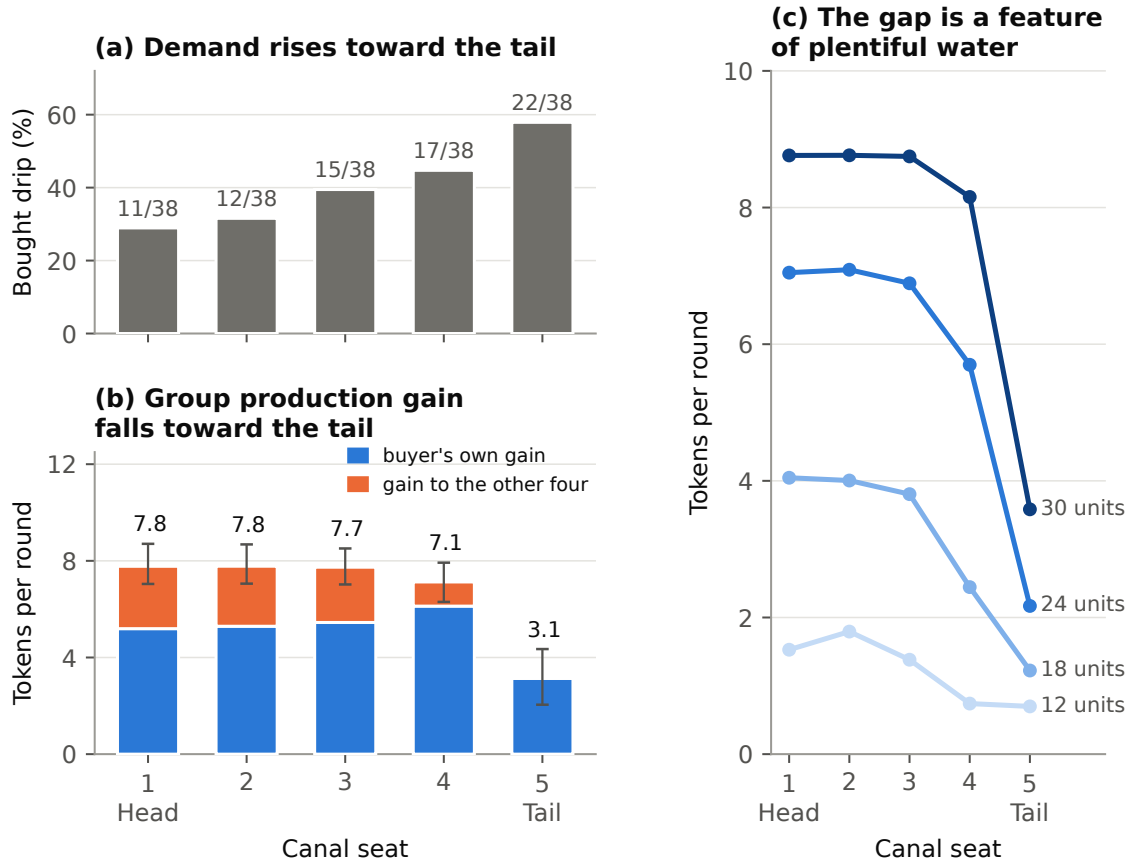


Figure 1: Demand and the group production gain run in opposite directions. Panel (a): share of farmers buying drip at the first decision of Session 2, by randomly assigned canal seat; labels give buyers out of 38. Panel (b): the group production gain from one installation at each seat in the canal model at Session 2 empirical supply, split into the buyer's own gain and the gain to the other four farmers; whiskers are 95 percent intervals from 199 game resamples that refit the canal rule. Panel (c): the total gain at constant supplies of 30, 24, 18 and 12 units. Gains are conditional on the calibrated canal rule; installments and the costs of exceeding a fair share are excluded.

Buying at the tail is a bet on the farmers above. Drip pays only if play lasts long enough and enough water arrives. Drip adds the most with a moderate amount of water, about what the tail receives when the four farmers above share. Each of them usually took close to a fair share, but all four did so together in only 38 percent of rounds. The tail received about 5 units of water when they did and 2 when they did not. Tail farmers bought mostly after a farmer above had bought drip. Of the tail farmers who had seen an upstream purchase, 21 of 28 bought, against 1 of 10 who

had seen none. Drip users take less water, so the private benchmark predicts this response, but it explains only about a third of it.

The private benchmark reproduces demand by seat (Figure 4), but not who buys. It explains only 5 percent of the variation in individual choices. The largest part, 41 percent, stays with the farmer after seats are reshuffled. It reflects each farmer's beliefs and attitudes and what the farmer learned in the first paid session. Farmers who use drip on their own farms, or whose own plot lies at the tail of a real canal, buy more. Survey measures together, however, explain little of this part.

A smaller part, 16 percent, is shared within groups. When we put farmers into random groups on paper, this shared part disappears. Farmers with a relative, neighbor or friend among the other players chose like the majority of the other four in 77 percent of decisions, against 55 percent for farmers without such a tie. Sharing water is more of a group behavior than buying drip (Section 5). But groups that shared water best in the practice session did not buy more. Cooperation in water use does not seem to move adoption toward the seats where it would help the group most.

Drip users upstream take less water, and more of it reaches the farmers below. With four drip users above, the tail had usable water in about 70 percent of rounds, against 40 percent with none. The farmers below use what arrives. In the canal model, they take up 95–96 percent of what an upstream adopter saves. Output rises, but total water use barely falls, even though no farmer can add land.

The findings add to studies of technology adoption, which explain who adopts by the return to the buyer and by learning from peers and experience [12, 22, 4, 9, 5]. In field data the return differs from farm to farm [28], so differences in returns are hard to tell from differences in beliefs or in neighbors. In our game everyone has the same public payoff cards, so we can tell them apart. Where a farmer sits on a shared canal changes demand for the same technology at the same price. Beyond the seat, who buys depends more on the farmer than on the group.

We also add a priced technology choice to experiments on irrigation commons [6, 15, 3, 2, 20]. In a canal game in South China, players could pay for a shared pump or for a weather forecast, and downstream players invested less [14]. Related

coordination experiments study investment in irrigation technology with positive network externalities [21], and an earlier framed game with Uzbek farmers compared rules for governing irrigation water [27]. We use randomized canal positions and a calibrated canal model to compare where farmers want the technology with where its group production gain is largest. Whether game behavior reflects the same participants' decisions outside the game is contested [16, 18, 31]. Farmers who report using drip on their own farms took about half a unit less water per turn in the practice session, before drip was available in the game (Section 7.3). Behavior in the game thus matches experience on real farms. The survey from which the players were recruited shows a pattern like the game's. Farmers who had never used drip were more likely to accept a subsidized offer of it when their plots lay at a canal tail, and drip goes with smaller yield gains on tail farms [19].

We provide controlled micro-level evidence on the irrigation efficiency paradox [13]. Studies of the paradox explain it by expanded irrigated area, thirstier crops and lost return flows. A model of the Upper Rio Grande basin shows that subsidies for drip can raise water depletion [32]. In Kansas, more efficient sprinklers raised groundwater extraction, partly through more irrigated land and thirstier crops [25], and in South India drip can raise profits without reducing water use [11]. In our game, land and crops are fixed, and extraction still barely falls. With 7 units or fewer, the most profitable use is all the water a farmer receives, on drip as on gravity. With 4 to 7 units, drip raises output while the output-maximizing withdrawal stays all the water received; with 3 units or fewer it adds nothing. Within a farmer, installed drip goes with no reduction in turns with fewer than 10 units. Saving water is not necessarily the privately preferred response to greater efficiency. Where water is plentiful, farmers who already share take less than the best use of drip, and the little they release is taken up downstream. We add where along a canal the saved water goes and which farmers ask for the technology.

The paradox also has a reason in who adopts. Drip is bought most readily as a remedy for scarcity, by farmers who are already short of water. For them it raises output with little or no reduction in withdrawals, and it releases little water for anyone else. It is also a risky bet. A tail buyer whose upstream farmers overuse

receives too little water for drip to pay and loses the installments already paid. Farmers with water to spare, whose adoption could release water, gain little and buy less often. Their incentive grows as supply falls far enough to constrain them, as at seat 4 in Session 3. In California, drip spread first where water was costly and fastest during droughts [29, 30]. Voluntary take-up may therefore concentrate where drip raises output with little reduction in withdrawals.

When a technology changes how a shared resource is divided, demand follows what the buyer expects to gain. At the price in our game, a program that relies on voluntary take-up may miss the places where adoption does the most for other farmers. The pattern gives a reason to consider and to test upstream installation combined with allocation rules that keep released water flowing to the farmers below. Programs should not count installations as water saved. Programs that measure delivered and consumed water are better placed to judge what technology achieves.

2 Setting and experimental design

Farmers along a shared canal make two decisions: how much to contribute to keeping water flowing, and how much to take when it arrives. Our experiment adds a third decision, whether to buy a technology that changes the return to the water received. It adapts the irrigation games of Cardenas, Rodriguez, and Johnson [6] and Janssen et al. [15] to the individual purchase of drip.

We recruited 190 farmers from a survey of 914 farms in Fergana, Kashkadarya, Samarkand and Khorazm. Recruiters telephoned farmers until five could attend together. The sample therefore reflects availability. The 38 groups kept the same five members throughout the experiment.

Survey records are linked for 142 participants; the other 48 have unknown survey status. We do not treat unlinked participants as farmers without drip. Appendix E compares linked participants with the survey. The games were played in December 2024: file timestamps date 14 groups to 4–20 December, and the Samarkand games fell in the middle of that campaign. Recorded survey starts span 30 October 2024–21 April 2025. Linked interviews were not uniformly pregame (Appendix A).

2.1 A shared canal and publicly specified payoffs

Each group represented a canal with five farms. Seats ran from the head, seat 1, to the tail, seat 5. Water taken upstream immediately reduced what remained for everyone below. Farmers extracted water in whole units, subject to the water available and the maximum of 13 units on the printed card. The gravity and drip cards specified production in tokens at each extraction level (Figure 2). The exact schedules appear in Appendix Table A2. Gravity production peaks at 40 tokens with 10 units; drip production peaks at 45 tokens with 7. The advantage of drip is greatest at moderate water use.

Each farmer received an endowment of 20 tokens every round, from which they could invest in canal maintenance. Total investment of 75 tokens, or 15 per farmer with equal contributions, maintained the infrastructure stock. Investment affected infrastructure in the following round; current extraction reduced current water. Supply depended on infrastructure and on a water cap set for each session. Appendix A gives the transition and supply equations. Maintaining the stock and maintaining delivered water are different when the cap binds.

Within each round, turns ran from upstream to downstream (Figure 3). Each farmer invested, decided whether to buy drip if eligible, and extracted before the next farmer acted. Every decision was visible to everyone. A farmer near the tail had seen more current decisions before acting. Farmers could talk between decisions and between sessions, but not while someone was deciding. Printed production payoffs were public; future withdrawals and the duration of play were uncertain.

2.2 Sessions, purchases and payment

All groups played the same sequence. Session 1 was unpaid practice to learn the gravity-only rules. Session 2 introduced drip with a water cap of 30 units. Session 3 offered drip again while the announced cap fell by two units per round, starting at 30. Seats were randomly reassigned between sessions. Drip expired between sessions, so repurchase required a fresh installation.

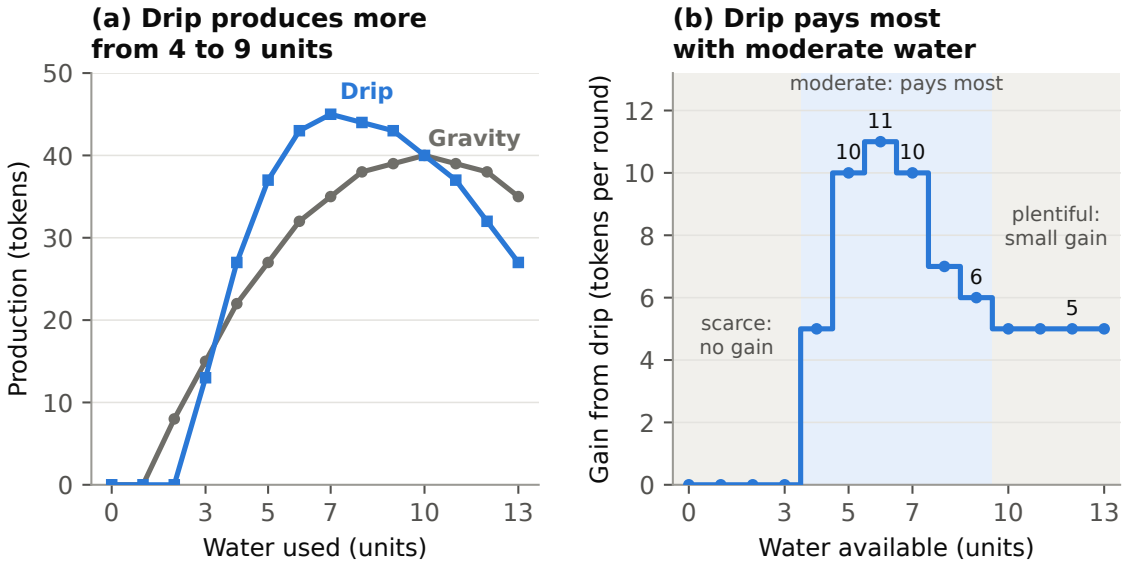


Figure 2: The printed production schedules and drip's gain. Panel (a) plots both cards at every allowed extraction. Panel (b) compares best feasible production at each water availability, retaining gravity as an option after installation. Shading marks scarce, moderate and plentiful water. Gains precede installments.

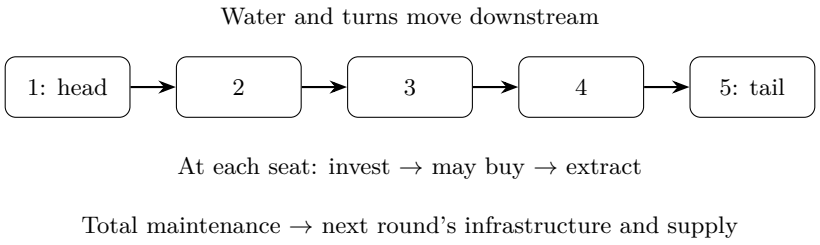


Figure 3: Canal and timing. A withdrawal reduces current water below that seat; maintenance changes the next round's supply. All five farmers observe each decision.

The endpoint was never announced. The protocol describes the number of rounds as determined by chance, with a 30 percent stopping probability (study protocol, paragraph 23). In the field, each session had a minimum length: 3 rounds in Session 1, 5 in Session 2 and 6 in Session 3. After the minimum, the facilitator's software drew a random end after each round with a probability set by the facilitator (0.3 by default). Farmers did not know the rule. After each session the facilitator told farmers only that the next would be longer.

In a few sessions the facilitator ended play before the minimum by procedural error: three Session 2 games ended at round 4, and two Session 3 games before round 6. Facilitators also sometimes ended play by hand. Practice could end when the facilitator moved on to the next game, and many Session 3 games were stopped around round ten to leave time for the post-game questionnaire. Realized mean lengths were 3.8 rounds in Session 1, 5.6 in Session 2 and 8.4 in Session 3.

A purchase started three installments of 15 tokens in its first three rounds of use. Installments still due when a session ended were not charged. Each installment entered the round's token account after investment and production, without reducing the investment endowment. Drip was available immediately. The rules allowed an installed farmer to use the gravity card in any round, including rounds with an installment due.

Negative payouts were floored at zero. The performance payment used one round drawn from the combined realized rounds of the two paid sessions. Farmers also received an attendance payment. Appendix A reports the payment units. In whichever round is drawn for payment, a purchase changes the account of that round by between -15 and $+11$ tokens.

The randomized comparison across seats identifies the joint consequences of position: water access, turn order and visible upstream decisions. A farmer's own purchase and upstream purchases were not randomly assigned. Farmers chose them. We distinguish the first decision of each paid session from purchase at any later point, and separate Session 3 entry by previous nonbuyers from repurchase by previous buyers.

2.3 Maintenance cooperation and unequal delivery

Tail farmers keep contributing to maintenance while extracting much less water than farmers at the head (Appendix Figure B1). Later-round means describe the groups still playing, whose numbers the figure shows. Cooperation in keeping the canal supplied coexists with unequal delivery.

3 Position and demand for drip

The allocation of water decides who wants drip, whether it pays and what it achieves for the farmers below. The cards and the price are the same for everyone, while the water a farmer can expect depends on the seat. We begin with who wants drip.

3.1 Demand rises toward the tail on first exposure

When drip first became available, farmers at the tail were twice as likely to buy as farmers at the head. At their first decision in Session 2, 11 of 38 head farmers bought (29 percent), against 22 of 38 tail farmers (58 percent). Farmers in seat 2 bought at a rate of 32 percent, in seat 3 at 39 percent and in seat 4 at 45 percent. The rise also held over the full session. Eventual purchases increased from 14 of 38 at the head (37 percent) to 28 of 38 at the tail (74 percent).

Random assignment makes the rise toward the tail a causal effect of canal position. Position combines physical access with a later turn and more decisions observed. The estimated linear slope is 7.1 percentage points per seat. An exact two-sided randomization test, permuting seats within each group and including tied statistics, gives $p = 0.002$. Appendix D reports the randomization calculation.

Most purchases were immediate. Across both paid sessions, 169 of 210 purchases occurred at the first decision: 77 of 97 in Session 2 and 92 of 113 in Session 3. Later purchases matter for eventual take-up, but they cannot account for the initial gradient by position. Figure 4(b) shows the first-decision rates, and Figure 6(a) separates Session 3 entry from repurchase.

Session 3 has a different profile. Its pooled first-decision rates are 47, 42, 50, 55 and 47 percent from head to tail. The pooled rates combine farmers making their first

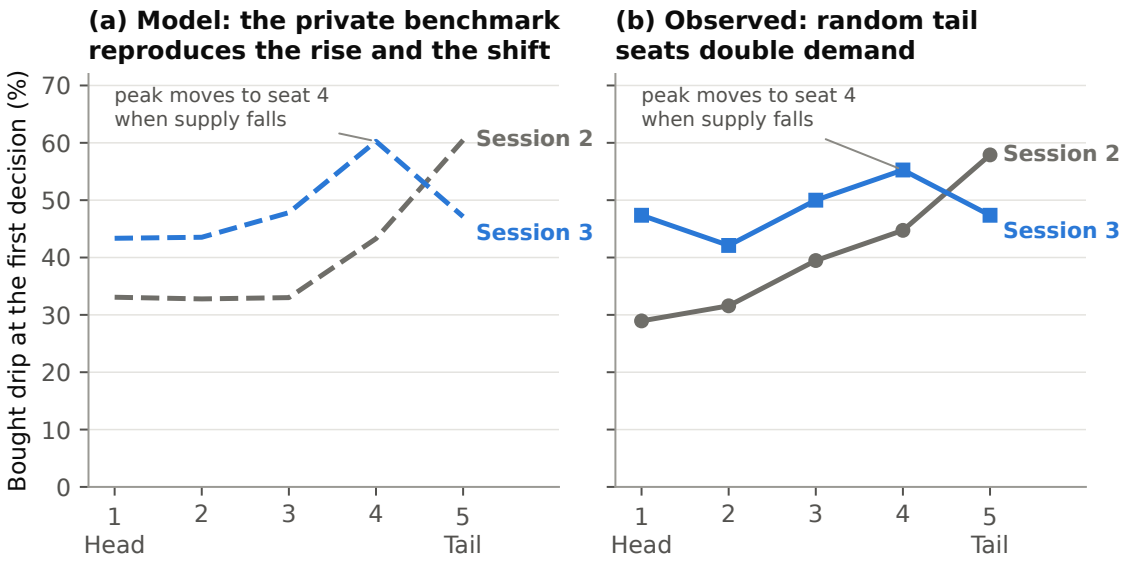


Figure 4: The private benchmark reproduces demand by seat. Panel (a): share of farmers predicted to buy drip at the first decision of each paid session, by canal seat, from the private benchmark of Section 4. Its beliefs are common to all farmers and fitted to each session, and it has no seat-specific parameters. Panel (b): observed share, with 38 farmers per seat and session. Seats were assigned at random. When supply fell in Session 3, the peak moved to seat 4 in both panels. Figure 6(a) separates Session 3 repurchase from entry.

purchase with farmers buying again after drip expired. Entry by previous nonbuyers is 5/19, 3/19, 3/15, 5/20 and 2/20 across seats. Repurchase is 13/19, 13/19, 16/23, 16/18 and 16/18. The higher conditional purchase rates at the downstream end in Session 3 are in repurchase. Entry has no rising gradient. Section 5 returns to the distinction between entry and repurchase. The pooled seat means and the regional comparisons in Section 7 do not establish a U-shaped demand profile.

3.2 What the printed cards imply

Consider first what a farmer could expect when deciding. The printed cards set drip's gain by the water available (Figure 2(b)). With 3 units or fewer drip adds nothing; with 5 to 7 units it adds 10 or 11 tokens per round; with abundant water it adds 5. Drip costs three installments of 15 tokens, so buying pays only if play lasts long enough and the water arrives in the useful range. A head farmer always had abundant water at the first decision. A tail farmer could gain most, but only if the farmers above left moderate water.

We then value drip on the realized paths, without fitting an adoption model. At each farmer's realized water and investment, we compare the best gravity extraction

with the best extraction available after installing. We deduct installments in the first three rounds and apply the payout floor separately to each alternative. We then sum the difference over the realized session. The calculation holds water and investment fixed. It answers what a purchase at the first decision would have paid on the realized path. Downstream water on the realized path already reflects actual upstream choices.

Table 1: Card value of buying at the first decision on the realized path

Canal seat	1	2	3	4	5
Session 2: mean length 5.6 rounds					
Mean net tokens	−17.6	−18.0	−19.1	−14.3	−10.2
Positive in groups, out of 38	1	0	0	2	6
Session 3: mean length 8.4 rounds					
Mean net tokens	−4.7	−5.0	−2.2	−2.0	−14.1
Positive in groups, out of 38	13	14	15	16	10

Notes: Each cell averages 38 farmer–session paths. A hypothetical initial purchase pays three installments of 15 tokens, with best feasible extraction on either card and a zero floor on each round's payout. The comparison uses actual investment and water at the farmer's seat. These are cumulative recorded-token differences, not estimated beliefs, expected cash payments or equilibrium counterfactuals. A positive group count requires a strictly positive net value.

Buying lost money on average at every seat in both sessions (Table 1). The profile nevertheless differs between them. In Session 2, the floor protects some installment rounds with little water, so the mean loss at the tail is smaller. In Session 3, the longer realized duration brings upstream losses nearer zero. Falling water makes the loss at the tail larger. The calculation therefore gives no common ranking of seats across sessions that tracks their purchases. It also shows why any valuation of drip must keep the installment schedule, payout floor and continuing gravity option.

Realized losses do not settle the decision facing a farmer who did not know when play would end. Section 4 develops the bet. The empirical question is how much of the purchase pattern the value measures we tested explain, under stated assumptions about water and duration.

The value measures we tested do not explain the full purchase pattern. Appendix Table C1 lists the accounts we tested, judged against the combination of downstream first purchases, early timing and repurchase, and Appendix C states their assumptions.

The cooperation counterexample (Appendix Table C1), a constructed case in which upstream farmers reliably leave moderate water, matters because it provides

an ordinary economic reason for downstream farmers to buy immediately. Farmers can expect moderate water to persist and enough rounds to recover the installments.

4 The private benchmark: a bet on the horizon and on upstream water sharing

Section 3 showed that canal position doubles demand for drip, but that the value measures tested do not explain the full purchase pattern. The allocation of water creates an investment problem that differs by seat, and a private benchmark describes it. The game gives every farmer the same payoff cards, rules and price, and it assigns seats at random. Water and what a farmer sees still differ by seat, and that difference is the treatment. A model of a farmer who maximizes their own expected pay can therefore serve as one private benchmark for everyone in a given seat and situation. The benchmark describes common strategy: what any farmer would do there, given how long play may last and what the others leave. Purchases that the benchmark does not explain point to the beliefs of particular farmers, to the behavior of their groups, or to what the model misses (Section 5). In the field the three are mixed with differences in soil, markets and skill, which change the return to drip itself. Buying drip is a bet. For every farmer it is a bet that play will continue long enough. For a farmer near the tail it is also a bet that the farmers upstream will leave enough water. The second bet explains why downstream farmers buy more, and why their purchases depend so strongly on what happens upstream. Upstream drip users take less water, so responding to their purchases is part of the private benchmark at the tail. The model shows when downstream purchases are sensible and how their returns depend on upstream behavior (Section 4.5).

4.1 One framework for extraction and adoption

We put extraction and purchase decisions in one framework. In each round, farmer i at seat k finds A_{ik} units of water. They extract c units and produce $y_j(c)$ on card j. The gravity card is always available, and the drip card is available once the farmer has installed drip. As a representation of observed behavior, we write the extraction

choice as

$$\max_{c \le \min(A_{ik},13)} \; y_j(c) - \kappa_i \max\!\left(0,\; c - \frac{A_{ik}}{6-k}\right), \tag{1}$$

where $A_{ik}/(6-k)$ is a fair share of the water left and κ_i is the cost the farmer attaches to exceeding it. The experiment does not identify whether the cost reflects fairness, social pressure or another motive. The water left for the next farmer is $A_{ik} - c$. A purchase is worthwhile when its expected gain, the increase in the maximized value of (1), is large enough for the length of play the farmer expects (Section 4.2).

The framework makes two points explicit. First, a downstream farmer's gain depends on how everyone else shares water, because the gain is evaluated at the water the others leave. The dependence makes buying at the tail a bet on upstream water sharing.

Second, how much a farmer holds back changes what drip is worth to them. At the head with 30 units, the fair share is 6. A farmer who does not hold back takes 10 units on gravity and 7 on drip, and drip adds 5 tokens per round. A farmer who holds back to 6 units on either card gains 11 tokens (43 against 32). Keeping to the fair share costs 8 tokens on gravity (40 against 32) but only 2 on drip (45 against 43). With plentiful water, drip makes sharing cheap. Cheap sharing is why an upstream installation can matter for everyone below (Section 6.5). With moderate water the reverse holds. At a seat with 8 units and a fair share of 4, holding back costs 16 tokens on gravity (38 against 22) and 18 on drip (45 against 27). Holding back costs more on drip because drip makes each unit up to 7 especially valuable. Drip can make holding back cheaper where water is plentiful and costlier where it is moderate, so drip and sharing can be complements in the first case and substitutes in the second.

How farmers extract also depends on how strongly they value sharing. In the data, farmers held back strongly: gravity users with 10 or more units available took 6.2 on average. Upstream purchases, however, look like the 5-token valuation. The framework explains the downstream bet (Sections 4.2–4.5).

4.2 The bet and the horizon

Drip changes what a farmer can earn from a given amount of water (Figure 2(b)). An installed farmer can still use the gravity card, so drip never lowers production in any round. The gain is zero with 3 units or fewer, largest with 5–7 units (10–11 tokens) and 5 tokens with abundant water.

Farmers pay for drip in three installments of 15 tokens over its first three rounds of use. Installments still due when a session ends are not charged. A farmer whose gain is e tokens per round falls $15 - e$ tokens behind in each of the three rounds, and recovers the 45 tokens of installments after $45/e$ rounds of use. With abundant water ($e = 5$) recovery takes 9 rounds; with 6 units ($e = 11$), about 4.

Whether drip is worth buying depends on how long play continues and on how pay is determined. Performance pay was one round drawn from all rounds of the two paid sessions. A purchase in session s therefore pays if it raises the expected account of the paid round, following the payment-scheme distinction of Chandrasekhar and Xandri [8],

$$\Delta V = \mathrm{E}\left[\frac{e\,T_s - 15\min(T_s, 3)}{T_s + T_o}\right], \tag{2}$$

where T_s is the length of the current session and T_o that of the other paid session. No time preference is involved, since all pay came at the end of the day. What matters is the length of play. A farmer who believes that each round is followed by another with probability 0.9, and so expects about 10 rounds, needs a gain of about 6 tokens per round for drip to pay. Expecting 20 rounds lowers the requirement to about 3.5 tokens.

Farmers were never told when a session would end. With realized joint session lengths as a descriptive benchmark, the requirement was 8.2 tokens for a purchase at the start of Session 2 and 5.6 at the start of Session 3. (The calculations abstract from rounds in which the payout floor binds; Table 1 applies the floor exactly.) The stopping rule of the protocol, a minimum number of rounds followed by a 30 percent chance of ending after each round, implies about 7.3 rounds for Session 2 and 8.3 for Session 3. Session 3 lasted 8.4 rounds on average, as the rule implies. Session 2 lasted 5.6, because some games were stopped early (Section 2). Farmers had no

reason to revise their horizon down after Session 2. They were told that Session 3 would be longer, and it was. Over the sessions as they unfolded, buying lost tokens on average at every seat (Table 1). Whether drip pays is therefore a question of expectations about both the horizon and the water, and the water a farmer receives depends on what other farmers do. Expected returns are themselves social.

For the three farmers nearest the head, the initial bet involves only the horizon. At their first decision they always had 10 or more units available, so drip added 5 tokens per round regardless of what anyone else did, and paid after about nine rounds of use. (Later in a session, especially under Session 3's falling cap, their water need not stay abundant.) Their initial purchases are then consistent with sufficiently optimistic expectations about the length of play. Under the paid-round rule, the required expectation depends on the whole distribution of session lengths and on the length of the other session, not on nine rounds alone. A third of them (38 of 114) bought at the first decision in Session 2. Farmers were told that Session 2 would be longer than the practice session, which had lasted 3.8 rounds on average, but not how long. Expectations too short to justify buying were reasonable. We do not observe horizon expectations directly, so we offer an interpretation of the upstream level, not an estimate. The interpretation implies that upstream non-adoption needs no special explanation. What needs explaining is why farmers farther down buy more.

4.3 The tail: a bet on upstream water sharing

The water at the tail is whatever the four farmers above it leave. Upstream farmers largely follow a norm of taking a fair share of the water left. A fair share is the water that reaches a farmer, divided equally among that farmer and everyone below. Two-thirds of upstream extraction decisions in the paid sessions were within one unit of a fair share, and the median decision exceeded it by half a unit.

The norm is not enforced through the infrastructure that all five farmers maintain:

- upstream extraction does not respond to how much downstream farmers invested in the previous round (coefficient -0.02, 95% interval -0.05 to 0.01);

- downstream investment does not respond to how much water a farmer received (coefficient −0.01, interval −0.08 to 0.05);
- farmers at every seat, the tail included, contribute close to the 15 tokens per round that the rules describe as maintaining supply;
- sharing close to a fair share is present even in the unpaid practice session.

Sharing is therefore consistent with a norm. It does not look like an exchange. The two regressions detect no sanction of overuse. In most of Uzbekistan, farmers clean and repair their canals together through hashar, unpaid communal work, in time for the spring planting season [1, 10]. Cooperative maintenance in the game is therefore not surprising (Appendix D.5).

An unenforced norm that is kept only approximately produces shortfalls at the tail without any clear culprit. All four upstream farmers stayed within one unit of a fair share of the water left in 38 percent of paid rounds. When they did, the tail typically received 5 units, where drip pays back fastest. When at least one took more, the tail typically received 2 units, where drip is worthless. In rounds that left the tail with 3 units or fewer, two or more upstream farmers had taken more than an equal share in 72 percent of cases. Farmers below an overuser offset only about an eighth of the excess, so nearly all of it passed downstream. If each had taken a fair share of the water left, they would have offset a quarter to a half. Each upstream farmer stays close to fair. The tail loses the sum of the excesses.

4.4 A simulation of the canal

To see how fragile the tail's bet is, we calibrate a canal model on both paid sessions and simulate it. (The out-of-sample check in Section 4.7 recalibrates it on Session 2 alone.) Each upstream farmer takes a fair share of the water left, plus three terms:

- an average excess that depends on whether the farmer has drip and how much water is available;
- a response to the excess already taken above them;
- an idiosyncratic deviation drawn from the observed residuals.

With 7 or more units available, farmers on gravity exceed a fair share by 0.7–0.9 units on average and drip users by 0.25–0.4. The response to excess taken above is +0.27 per unit: farmers below an overuser take somewhat more than a fair share of the water left, not less. The canal model reproduces the observed water at the tail reasonably well: the tail receives 4 or more units in 46 percent of simulated rounds, against 52 percent observed. It is somewhat too optimistic when calibrated on Session 2 and applied to the scarcer Session 3.

Table 2: The tail's bet in the simulated canal (supply of 30 units)

Upstream behavior	Tail receives 4+ units	Tail receives 5–7 units	Expected gain per round	Rounds of use to break even
Exact fair shares	100%	100%	11.0	4.1
Calibrated, no upstream drip users	50%	30%	4.2	10.7
Calibrated, two upstream drip users	65%	43%	5.7	7.9
Calibrated, four upstream drip users	83%	60%	7.5	6.0
Head farmer, for comparison	—	—	5.0	9.0

Notes: 20,000 simulated rounds per row. Break-even rounds are undiscounted (45/gain). Adding upstream drip users means having more upstream farmers who behave as drip users do.

Table 2 gives the result. If upstream farmers took exactly a fair share, the tail would receive 6 units and drip would pay back in about four rounds, sooner than at any other seat. Under the calibrated behavior with no upstream drip users, the tail receives usable water only half the time, and its bet is worse than the head's. The bet is also fragile. If every upstream farmer took half a unit more than calibrated, the tail would receive usable water in only 26 percent of rounds. Its expected gain (2.1 tokens) would then fall below what drip needs to pay even for a farmer expecting 20 rounds of play, about 3.5 tokens (Figure 5).

4.5 What the private benchmark explains, and what it leaves

Upstream drip users overuse less than farmers on gravity. With abundant water, 17 percent of drip users took more than a unit above a fair share of the water left, against 40 percent of gravity users. Drip's best use is 7 units, against 10 on gravity, which may itself curb overuse. How much a farmer held back in the practice session does not predict who buys, and the change for the same farmer after installing is imprecisely estimated. We therefore cannot say whether drip changes behavior or farmers who share more choose it. For a farmer downstream the distinction does not

matter. In rounds with no upstream drip users, the tail received usable water in 40 percent of cases; with four, in 71 percent. Seeing an upstream farmer buy is good news about the tail's bet.

The model therefore predicts an ordering at the tail. Without upstream adopters, the tail's bet is worse than the head's (Table 2), so tail farmers who have seen no upstream purchase should buy less than head farmers. With upstream adopters the bet is better, so tail farmers who have seen one should buy more. Purchases at the first decision in Session 2 follow the predicted order. Among tail farmers who saw no upstream purchase, 1 of 10 bought; among head farmers, 11 of 38; and among tail farmers who saw one, 21 of 28. We did not fit the canal model to these purchases.

We fit the benchmark with beliefs common to all farmers. In Session 2 they expect about 10 rounds of play, and they expect each upstream farmer to take about half a unit less than calibrated. With these beliefs the benchmark reproduces demand by seat (Figure 4(a) and Section 5). Seeing an upstream buyer then raises the tail's expected gain from 5.7 to 6.8 tokens per round and its predicted purchase rate by 20 percentage points. The observed rise is 65 points. In Session 3 the benchmark predicts 17 points, against 58 observed. The private benchmark predicts the direction of the tail's response and about a third of its size. Farmers differ in how long they expect play to last and in what they expect from others. Seeing an upstream purchase may raise a tail farmer's expected water, expected horizon or both. A group that adopts readily may move all of its members together (Section 5).

The bet explains the direction of the tail's swing. It does not explain the whole gradient. Because seats 1–3 have abundant water at the first decision, the model predicts the same initial incentive at all three. In the data, demand rises modestly across them, from 29 to 39 percent. Farmers at seats 2–4 also respond to an upstream purchase, by 19 to 31 percentage points, although their water hardly depends on it.

Upstream farmers usually held back to about 6 units. A farmer who expected to keep doing so would value drip at 11 tokens per round (43 against 32 on the printed cards), more than anyone downstream, and should buy most often. Upstream farmers bought least. Their rate of about a third fits valuing drip against what they could take, 5 tokens per round. The same fairness that limits their extraction might

be expected to enter their valuation too. Their own sharing enters only weakly. We measure how much each upstream farmer shared before the decision, in the practice session or in Session 2. Those in the third who took the most beyond a fair share bought at 29 percent of their first decisions, and the other two thirds at 41 percent. The top half against the bottom half is 45 against 30 percent (permutation $p = 0.04$), although a linear fit is weak ($p = 0.46$). Under the benchmark, an 11-token valuation would make nearly every sharer buy. Farmers appear not to count fully on their own future sharing when they invest, which fits sharing being more of a group behavior than a personal plan (Section 5). In the utility of equation (1), the gain from drip lies between 5 and 11 tokens and rises with the cost a farmer attaches to exceeding a fair share. Farmers may also compare the peaks of the two printed cards, 45 against 40 tokens, which the cards display most plainly. The data cannot fully settle which comparison they made.

The model explains why downstream purchases are sensible bets and why they depend on what happens upstream. Four patterns remain: most of the tail's response to upstream buyers, the rise across seats 1–3, the response of seats 2–4 to visible purchases, and the persistence of purchases across sessions. They cannot come from differences in the printed payoff cards or the price, and they remain after accounting for the common private benchmark. They belong to particular farmers and to their groups, and Section 5 measures how much of who buys each part accounts for.

4.6 One payment rule, regional beliefs

The regions differ in their purchase profiles (Table 10). In Session 2 the tail bought most in Fergana, Kashkadarya and Samarkand. In Samarkand all 8 tail farmers bought. In Khorazm the head bought more than the tail, 5 of 9 against 2 of 9. In an exploratory, as-if description, every farmer values a purchase by equation (2) with one choice-noise scale. Each region holds its own beliefs about the continuation of play and about upstream water sharing, fitted to the 190 Session 2 first decisions. Horizon beliefs come out similar across regions (10–12 expected rounds), so the regional differences fall on beliefs about upstream water sharing. Without upstream adopters, Samarkand's farmers act as if the tail would receive 5.7 units and Khorazm's as

if it would receive 2.8. The as-if beliefs run opposite to what the games in each region delivered. Appendix C reports the estimates, intervals and regional Session 3 comparisons. With four regions and 8–11 farmers per region and seat, we read the estimates ordinally.

4.7 Session 3 out of sample

Session 3 changed two conditions. Play lasted longer (on average 3.8 rounds in the practice session, 5.6 in Session 2 and 8.4 in Session 3). The cap also fell by two units per round, so the average supply per round dropped from 27.1 to 20.7 units. We hold the canal model and the regional beliefs at their Session 2 values and apply them to Session 3's supply and the realized Session 2 length of each group. We refit nothing, so Session 3 is out of sample (Table 3).

Table 3: Session 3 out of sample: incentive and purchases by seat

Canal seat	1	2	3	4	5
Rounds of use to recover the installments					
Session 2	9.0	9.1	9.1	7.3–8.4	6.4–11.1
Session 3	9.4	9.2–9.3	8.4–8.7	7.0–7.4	8.5–15.0
Session 3 purchases at the first decision (%)					
Predicted, Session 2 beliefs	28	29	39	68	43
Observed, all farmers	47	42	50	55	47
Observed, first-time buyers	26	16	20	25	10
Observed, previous buyers	68	68	70	89	89
Session 2 observed, for comparison	29	32	39	45	58

Notes: Rounds of use are 45/gain, with the expected gain simulated from the canal rule calibrated on Session 2 at each session's empirical supply. Ranges span the cases of all upstream farmers on gravity and all on drip. Lower values mean a stronger incentive. Predicted rates use the regional beliefs of Section 4.6, fitted on Session 2. Each seat has 38 farmers per session; first-time buyers and previous buyers are split by Session 2 purchase.

The model makes three ordinal predictions. First, at seats 1–3 water stays abundant, so their incentive barely changes (8.4–9.4 rounds). With beliefs held at their Session 2 values, the model predicts little change there. It underpredicts: purchases rose at every seat from 1 to 4, mostly through repurchase. The gap is what longer expected play would produce. Farmers had just played a longer session and were told that the next would be longer still.

Second, with less water the strongest incentive moves from the tail to seat 4, which now receives water in drip's most valuable range. The model predicts the highest rate at seat 4, and seat 4 had the highest observed rate, 55 percent. Seat 4

was also the only seat where buying at the start of Session 3 had positive expected value under the realized-length benchmark. At the start of Session 2, no seat did.

Third, without upstream drip users the tail's bet becomes the worst on the canal, at 15 rounds. The tail had the fewest first-time buyers, 2 of 20. The model misses at the head, where 5 of 19 previous nonbuyers bought, more than the ranking implies. The cells are small. The model also does not explain why 16 of 18 previous buyers bought again at the tail. Section 5 shows that repurchase followed reliable water.

4.8 The social optimum: everyone adopts and shares

A planner who maximizes group production would put every farmer on drip. The planner's choice is a benchmark constructed from the printed cards, not an estimate of what coordinated adoption would yield on real farms. With the full supply of 30 units and five farmers on gravity, the best allocation gives each farmer 6 units, for group production of 160 tokens per round. With all five on drip, the same equal allocation yields 215 tokens. Each adopter adds 11 tokens per round, and the group recovers its 225 tokens of installments in about four rounds, less than the average length of Session 2. Conditional on full supply and equal sharing, universal adoption therefore pays within the average realized Session 2.

The gain comes from producing more with the same water, not from saving water: every efficient allocation uses all 30 units. When water is scarcer the benchmark changes. With 20 units, the planner would put four farmers on drip and give the fifth no water at all. Universal adoption is therefore efficient only when every farmer has enough water to use drip well.

The benchmark is universal adoption, but only 40 percent bought at the first decision. Drip creates the efficient outcome only together with equal sharing. At an equal share of 6 units, drip is worth 11 tokens per round to every farmer, and it pays back in about four rounds wherever they sit. Upstream farmers, however, appear to value drip against what they could take, which gives 5 tokens and nine rounds. The tail can capture its share of the gain only if everyone above it keeps to the norm. Universal adoption requires both technology and allocation, and farmers can buy only the first.

The claims of the model are orderings. The tail buys less than the head without an upstream buyer and more with one. When water falls, the strongest incentive shifts to seat 4, and seat 4 buys most. The regions differ in their as-if beliefs. The model explains why downstream purchases are bets on the behavior of others. It shows that the downstream purchases that looked unprofitable are bets that pay when upstream farmers keep to a fair share. Sharing within one unit of a fair share is common for each farmer but holds for all four at once in only 38 percent of paid rounds.

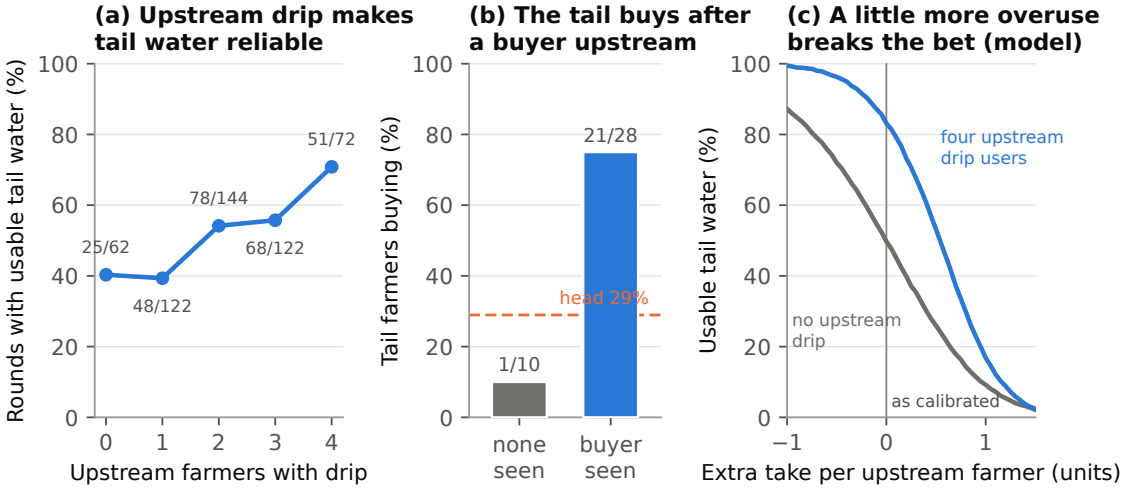


Figure 5: The tail's bet. Panel (a): share of complete paid rounds in which the tail received at least 4 units, by the number of upstream farmers with drip installed; labels give rounds. Panel (b): share of tail farmers buying at the first decision of Session 2, by whether an upstream farmer had bought before them; the dashed line is the head's rate. Panel (c): in the canal model at a supply of 30 units, the probability that the tail receives at least 4 units as each upstream farmer takes more beyond a fair share than calibrated (0 is the calibrated rule), with no upstream drip users and with four; 20,000 simulated rounds per point. Panels (a) and (b) are observed; panel (c) is a model calculation.

5 Beyond the private benchmark: personal and group parts of adoption

Buying at the tail can be a reasonable bet on future water. Farmers respond to the investment problem through their own beliefs and through their groups, and the benchmark leaves room for both. The private benchmark of Section 4 describes what any farmer would do in a given seat and situation. The game fixes the cards, the rules and the price, so what the benchmark leaves cannot come from differences between farms in soil, markets or the technology. It comes from the farmers themselves, from

their groups, from what the model misses, or from chance. Purchases persist after drip expires, and farmers in the same group often make the same choice. Farmers at middle seats also buy more after a visible upstream purchase, even when their own water is abundant. We first measure how much of who buys each part accounts for, and then examine the personal and the group parts in turn. We describe the patterns using first decisions, so ownership accumulated during a session does not enter the outcome. Figure 6 summarizes the three main patterns.

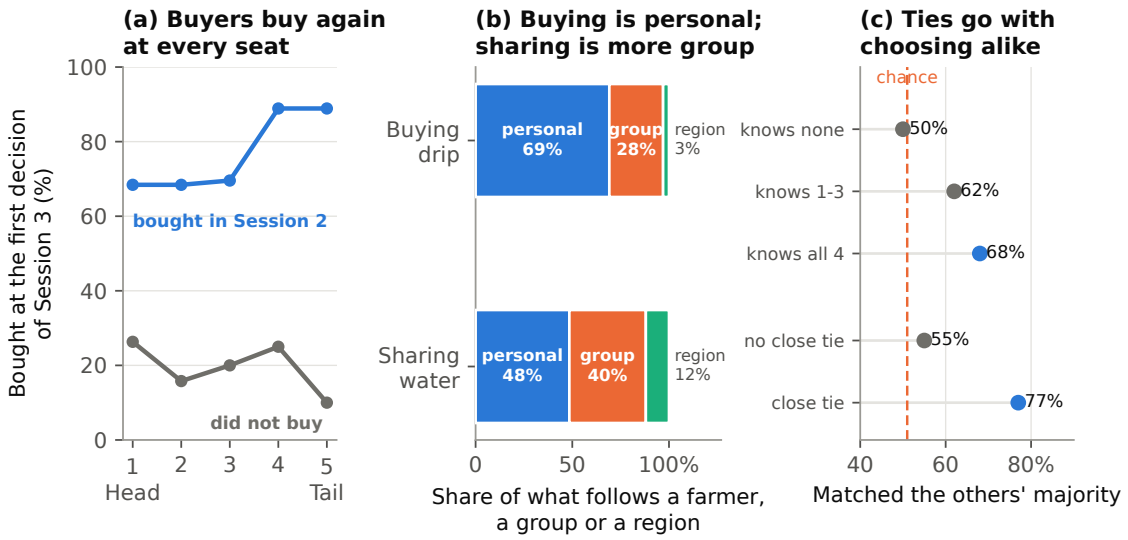


Figure 6: Who buys beyond the private benchmark. Panel (a): share buying at the first decision of Session 3, by seat, for farmers who did and did not buy in Session 2; seats were reassigned between sessions. Panel (b): of the variance that follows a farmer, a group or a region, the personal, group and regional shares, for first-decision purchases and for each farmer's mean excess over a fair share in a session (Table 5). Panel (c): share of decisions in which a farmer matched the majority choice of the other four, by the farmer's ties to them; a close tie is a relative, neighbor or friend among the players, and the dashed line is what independent choices would produce. All panels are associations.

5.1 The benchmark reproduces demand by seat but not who buys

The private benchmark reproduces the profile of demand across seats (Figure 4). With beliefs common to all farmers, it predicts first-decision purchase rates of 33, 33, 33, 43 and 60 percent from head to tail in Session 2, against 29, 32, 39, 45 and 58 percent observed. In Session 3 it predicts 43, 44, 48, 60 and 47 percent, against 47, 42, 50, 55 and 47 percent.

The benchmark leaves most of the variation in individual choices unexplained. At any seat it predicts a rate between about 30 and 60 percent, so some farmers buy and some do not. Table 4 divides the variance of the 380 first decisions in Sessions 2 and

3. The common part, seat and session together with the response of the benchmark to visible upstream purchases, accounts for 5 percent. We divide the rest by the covariances between pairs of decisions. Two decisions by the same farmer, made at different seats, share a personal part. Decisions by different farmers in the same group share a group part, within a session or across both. Decisions in the same region share regional beliefs.

Table 4: Who buys: common strategy, personal and group parts of first-decision purchases

	Common part measured by	
Share of the variance of purchases	Position only	Private benchmark
Common strategy	0.03 (0.02)	0.05 (0.02)
Regional beliefs	0.02 (0.03)	0.02 (0.02)
Group, within a session	0.15 (0.06)	0.12 (0.06)
Group, across sessions	0.06 (0.09)	0.04 (0.08)
Personal	0.40 (0.07)	0.41 (0.08)
Not shared with any other decision	0.37 (0.05)	0.38 (0.05)

Notes: 380 first decisions in Sessions 2 and 3 by 190 farmers in 38 groups; the variance of purchase is 0.247. Position only: seat-by-session purchase rates. Private benchmark: the same rates plus the response of the payment-rule model, with beliefs common to all farmers, to the upstream purchases visible at the decision. Each part is a covariance between pairs of residual decisions, measured against unrelated pairs in the same region (Appendix D.5). Jackknife standard errors by group in parentheses. The rows sum to one up to that reference covariance (-0.02). All estimates are associations.

The personal part is the largest, at 41 percent of the variance. Shuffling the Session 3 decisions among the members of each group removes it (permutation $p < 0.001$). The group part adds 16 percent, most of it within a session. When farmers are regrouped at random within their region, the group part falls to 0.3 percent ($p < 0.001$). Regional beliefs add 2 percent. The remaining 38 percent is not shared with any other decision we observe.

The personal part combines stable beliefs and attitudes with what each farmer learned from their own experience in Session 2. The design does not separate the two. Both follow the farmer and not the seat, and neither is a difference in the return to drip, which the cards hold fixed.

5.2 The personal part: farmers buy again after the reset

The personal part shows most directly in repurchase. At the first decision of Session 3, 74 of the 97 Session 2 buyers bought again, and 18 of the 93 previous nonbuyers entered, buying for the first time. The rates are 76 and 19 percent. The gap survives even though drip expires between sessions and seats are reassigned. It is persistence

in choice. Prior purchase was voluntary, so the comparison cannot separate an effect of having bought from differences between buyers and nonbuyers. Both follow the farmer and not the seat.

When later decisions also count, Session 3 repurchase rises to 81/97 and entry to 32/93, or 84 and 34 percent. The full-session rates answer a different timing question from the first-decision comparison. In both cases, previous buyers are much more likely than previous nonbuyers to buy. From here on, the outcome is again purchase at the first decision.

In Session 3, the purchase rate by seat pools repeat and first-time buyers, so it says little about new demand (Figure 6(a)). Repurchase rates rise from 68 percent at the head to 89 percent at the tail. Entry rates are 26 percent at the head and 10 percent at the tail. At the tail, the pooled rate of 47 percent comes from 16 repeat buyers and only 2 first-time buyers. Conditional on purchase history, the rise toward the tail is therefore in repurchase. It does not show that downstream farmers who previously declined became more willing to buy as water grew scarcer.

Repurchase followed reliable water access. The link is to water delivered while drip was installed, not to recovery of the installments. Only 6 of 97 buyers recovered them during Session 2. Session 2 buyers at seats 4 and 5 depended on upstream farmers for water. For them, the outcome that matters is whether usable water (4 or more units) reached them in the rounds they had drip. Of the 28 downstream buyers who received usable water in at least three-quarters of their drip rounds, 25 bought again (89 percent), against 12 of 22 who received it less often (55 percent). In a linear probability model with Session 3 seat controls, moving from no usable rounds to all usable rounds raises repurchase by 0.65 (95 percent interval 0.07 to 1.24, $n = 50$). The water a buyer received depended on how their own group shared water, and the same group may also shape their willingness to buy.

Field experience and real canal position line up with buying beyond the benchmark in the direction of the canal story (Appendix Table D6). Every linked farmer is kept: farmers whose main plot is not on a canal form their own category. Per decision, farmers whose own plot lies at the tail of a real canal buy 9 percentage points more than the benchmark. Farmers in the middle buy 3 points more and farmers at the

head 2 points less. Farmers off canals buy 3 points more. The 11 farmers who use drip on their own farms buy 19 points more. Members of agricultural clusters, farmers with larger farms, more risk-averse farmers and farmers who shared more in the practice session also buy more. The 11 women buy 31 points less. The signs hold with no controls, within region and within group, but each estimate rests on few farmers and is imprecise.

Experience in the game points the same way. Seats in the practice session were random, and sharing is undefined for the farmers who sat at the tail, since no one was below them. We keep them with an indicator. The 38 farmers who sat at the tail in practice buy 9 points more than the benchmark in the paid sessions, against 2 points less for the others, whatever their later seat. The estimate is imprecise (p between 0.13 and 0.18). It suggests that part of the personal part is learned from experience of scarce water.

Stated willingness to pay and beliefs about drip show no consistent link. Farmers who refuse drip at every subsidy in the survey buy more in the game, mostly in Khorazm (Section 7). Within groups, farmers who need a larger subsidy buy slightly less. Expected changes in water use and yield from drip, and the expected effect of wider adoption in the district, do not predict buying. In the same survey, these beliefs do not predict plans to adopt either.

The survey measures together account for a modest share of who buys. All of them explain 19 percent of the variance of a farmer's propensity in sample, and none of it in groups left out of the fit. They account for about a tenth of the group part beyond chance, as expected when groups are recruited locally and share field conditions. They account for almost none of the personal part.

The personal part is therefore mostly what field data do not reveal. It may reflect beliefs about how long play lasts and about other farmers, attitudes toward drip, or lessons from the first paid session. The common payoff cards, rules and price are what make it visible.

5.3 The group part: purchases cluster within groups

Groups often chose together when drip first appeared. Of the 38 groups in Session 2, 9 had no first-decision buyers and 2 had all five buying. The 11 unanimous groups compare with 3.1 expected if the same total number of buyers had been spread at random across groups. The permutation probability is below 0.0001. An exact calculation conditional on the total number of buyers gives the same conclusion. Appendix Table D2 reports the full distribution. The result concerns agreement in purchases, regardless of which farmer moved first.

Visible upstream purchases also predict own purchases. At seats 2–5 in Session 2, 48 of 80 farmers who had seen an upstream buyer bought, compared with 18 of 72 who had seen none: 60 versus 25 percent. At the tail the contrast is larger, 21/28 against 1/10. Section 4 explains why an upstream purchase can be good news about future water sharing. At middle seats, however, the association is present even though one more upstream adopter changes the private water incentive of farmers there only a little. The private benchmark makes the comparison exact. At seats 2–4 in Session 2, seeing an upstream buyer leaves the expected gain from drip at 5.2 tokens per round, and the benchmark predicts no change in buying. Observed buying rises from 27 to 52 percent. At the tail, the benchmark predicts a rise of 20 percentage points, against 65 observed (Section 4.5).

The raw contrast between farmers who saw an upstream buyer and those who did not largely reflects which groups adopt. Suppose we keep every farmer's first choice and purchase history but reorder the five farmers at random within each group and session. The coefficient on seeing an upstream buyer is then 0.143 on average, close to the observed 0.147. The permutation distribution puts the observed coefficient well inside its central range. Discussion between decisions, common views about drip and shared local conditions can all make a group adopt readily.

The reference coefficient for a regression with group effects also matters. In a small group, the upstream count excludes a farmer's own choice, but the group outcome used by the fixed effects includes it. The mismatch creates a negative reference coefficient even under random order. The first-decision specification gives

-0.037. Its reference is the -0.056 obtained under random order, not zero. The estimates show strong clustering, with limited evidence of an additional response to the order of visible choices. Appendix D gives both specifications and the permutation comparison.

The reordering pools seats 2–5. At most of them an upstream purchase changes a farmer's water little. At the tail an upstream purchase changes water most, and there purchases depend more on the choices of others than at the head. The head decides first and sees no one, so the link between its purchase and the choices of its group reflects clustering. A tail farmer is 65 percentage points more likely to buy when another farmer in the group bought. A head farmer is 38 points more likely. Permuting seats within groups gives a one-sided p of 0.04 for the difference. We made the check after seeing the data. The contrast arises in groups where someone else bought (the tail bought in 21 of 28 such groups, the head in 11 of 29). It is therefore consistent with the tail's water bet.

Clustering alone cannot make demand rise toward the tail. Seats are assigned at random, so every seat draws from the same mix of groups, and a group's readiness to buy is the same on average at the head and at the tail. A rising average needs either an incentive that differs by seat or herding that runs one way, in which a visible buyer triggers purchases and visible decliners deter no one. Herding of that kind fits the middle seats, which buy more after any upstream purchase. One-way herding depends only on the order of decisions, so it would keep the tail highest in every session. When supply fell in Session 3, demand peaked instead at seat 4, 55 percent against 47 at the tail, where the water incentive had moved. The tail's own response to upstream buyers fits the water bet, although alignment within the group can produce it too. The private benchmark, which contains no herding, also reproduces the profile of demand across seats (Section 5, first subsection).

Agreement in buying is stronger among farmers with social ties. After the games, each farmer said how many of the other four players they knew and how. When most of the other four made one choice, farmers who knew none of them matched it in 50 percent of decisions, close to what independent choices would produce. Farmers who knew some of them matched it in 62 percent, and farmers who knew all four in

68 percent (permutation $p = 0.05$ against those who knew none). Farmers with a relative, neighbor or friend among the players matched it in 77 percent, against 55 percent without such a tie ($p = 0.002$).

At the level of the group the relation is not linear (Appendix Table D7). In groups of mostly strangers, where members knew on average fewer than 1.5 of the others, 13 percent of sessions were unanimous at the first decision. In groups with some ties it was 55 percent ($p = 0.005$), and in close-knit groups, where members knew more than 3 of the others, 25 percent. We chose these categories after seeing the data. Groups with ties did not buy more on average. Ties go with choosing alike, in either direction.

Three quarters of the group part arises within a session, and agreement is stronger among farmers who know each other. The majority includes farmers who decided later, so matching it is not the same as following an observed choice. It points to discussion between decisions or to following the visible choices of people one knows. District field conditions and the composition of the group predict it no better than chance (Appendix Table D7 notes).

Appendix D summarizes the associations in a compact purchase logit. Previous purchase and upstream purchases remain strongly associated with buying. The coefficient on the expected gain from drip is positive but imprecise. The value measures tested do not explain the full purchase pattern.

5.4 Cooperation in sharing water does not carry into buying

Groups that shared water in the practice session later delivered more water to the tail. The same groups did not buy more drip. A cooperation index for the practice session combines extraction held to a fair share and infrastructure investment before drip is available. Its rank correlation with the share of the group buying at the first decision of Session 2 is only 0.05, with permutation $p = 0.75$. The index therefore measures a group trait that predicts later water outcomes. It does little to predict adoption.

The difference between sharing and buying matters for how we read clustering. Upstream adopters leave more usable water, and observing them can improve the

outlook of a downstream farmer. A group that buys together, however, need not be a group that shared particularly well beforehand. The associations with purchase history and with visible purchases describe a pattern of adoption that the water account alone leaves incomplete.

Individual sharing points the same way. Its link to buying is weak. Upstream farmers usually held back to about 6 units, close to a fair share. Valued at that extraction, drip would add 11 tokens per round, and upstream farmers should buy most often. They bought least, at a rate that fits valuing drip against what they could take, 5 tokens per round (Section 4.5). Farmers who shared more in the practice session buy only slightly more beyond position (Appendix Table D6).

The same decomposition applied to sharing shows the contrast directly (Table 5). For each farmer and session we take the mean excess over a fair share of the water left at seats 1–4. The common part is the mean excess of anyone at the same seat, with the same card and water. The group part of sharing is 22 percent of its variance, against 4 percent when farmers are regrouped at random ($p < 0.001$). Part of the group part of sharing may reflect conditions that a group shares within a session, such as how the supply falls from round to round. Of the part that follows a farmer, a group or a region, groups and regions account for 52 percent in sharing and 31 percent in buying. Buying is mostly personal; sharing is more of a group behavior.

Social ties go with closer matching in sharing, as they do in buying. Consider what happens when the other members of a group took more than a fair share. A farmer with a relative, neighbor or friend among them took 0.8 units more for each unit of their excess. A farmer without such a tie took 0.4 units more (permutation $p = 0.04$). A group's sharing and its buying are nearly unrelated: the rank correlation is +0.13 across the 38 groups and +0.07 across farmers.

Who shares is a stable trait that appears before drip exists (Appendix Table D8). A farmer who shared one standard deviation more in the practice session takes 0.29 units less beyond a fair share per decision in the paid sessions ($p < 0.01$ with no controls, within region and within group). Practice-session behavior accounts for 40 percent of the personal part of sharing, against 4 percent for placebo values shuffled across farmers. It is the only set of measures that predicts groups left out of the fit.

Table 5: Buying is mostly personal; sharing water is more of a group behavior

	Buying drip	Sharing water
Share of the variance		
Common strategy	0.05 (0.02)	0.18 (0.05)
Regional	0.02 (0.02)	0.07 (0.03)
Group, within a session	0.12 (0.06)	0.17 (0.06)
Group, across sessions	0.04 (0.08)	0.05 (0.08)
Personal	0.41 (0.08)	0.27 (0.08)
Not shared with any other decision	0.38 (0.05)	0.29 (0.09)
Share of the part that follows a farmer, a group or a region		
Personal	0.69	0.48
Group	0.28	0.40
Regional	0.03	0.12

Notes: Buying: the 380 first decisions of Sessions 2 and 3, with the private benchmark as the common part (Table 4). Sharing: each farmer's mean excess over a fair share of the water left in a session, at seats 1–4 of Sessions 2 and 3 (304 farmer-sessions, 6.9 decisions each on average); the tail is excluded because no one is below it. The common part of sharing is the mean excess at the same session, seat, card and water band; the response to excess taken above is left in the group part. Jackknife standard errors by group in parentheses. A buying decision is a single choice, while sharing averages several decisions, so the lower panel compares the parts that follow a farmer, a group or a region. For sharing, the group parts are 0.22 against 0.04 when farmers are regrouped at random within region ($p < 0.001$). Including the practice session gives group 0.20 and personal 0.18.

Life on a real canal points the same way. Farmers whose main plot is not on a canal, and who irrigate from wells, rivers or rain, take the most beyond a fair share, 0.26 units more per decision than the benchmark. Farmers at the head of a real canal take the least, 0.23 units less, and farmers in the middle and at the tail lie between. Farmers who know more of the other players and farmers with better water in the peak season also take less. The signs hold with no controls, within region and within group, but most estimates are imprecise. Willingness to pay and beliefs about drip show no clear pattern.

Farmers share water under a norm of taking a fair share, but the cooperation does not carry into buying. The group part of buying is clustering, and it is unrelated to how well the group shared water before drip existed. The patterns do not show how much weight farmers give to the gains of others. Cooperation in water use does not move adoption toward the seats where it would add most to group production, which Section 6.5 measures.

6 Where the released water goes, and who gains

Demand concentrates downstream, while the group production gain of an installation is largest upstream when water is plentiful (Figure 1). Panel (a) shows observed purchases under randomized seats; panels (b) and (c) show gains in the canal model.

Who adopts decides where released water goes and who gains from it, and the use of that water downstream decides how much is saved. The fall in total extraction equals the reduction by the adopter minus the extra extraction downstream. Efficiency can raise production without reducing total extraction. In the game, land and crops are fixed, so the usual explanations of the paradox, more irrigated area and thirstier crops, cannot apply. The paradox appears without any expansion. Two features of the cards and of play do the work instead. With 7 units or fewer, the most profitable use is all the water a farmer receives, on either card. With more water, farmers already hold back close to a fair share, and the water they do release flows to the farmers below. The cost of exceeding a fair share in equation (1) explains why comparing the private optimum on each card need not describe the water released in play.

6.1 Adopters release little water in play

For the same farmer, installed drip is associated with 0.28 fewer units extracted per round, with a game-clustered standard error of 0.15. The regression controls for the water available, session and round, and excludes infeasible extraction choices. The reduction is concentrated in turns with at least 10 units available: 0.42 units, with a standard error of 0.24. Below that threshold, the estimate is close to zero. Farmers chose whether to install, so the within-farmer comparison and the others below that rest on observed installations are associations.

The small reduction is consistent with how farmers were already sharing. Gravity users with 10 or more units available took an average of 6.2 units, and only 22 percent took at least 8. On drip, a farmer's own output is highest at 7 units. Moving every gravity user with 10 or more units to 7 units would therefore raise their average extraction. The engineering comparison between card optima, 10 units for gravity and 7 for drip, overstates the reduction available when gravity users already hold back. Existing cooperation changes the baseline against which a new technology should be judged. Table 6 separates raw means by water availability from the within-farmer comparison.

Table 6: Observed extraction and installation status

	Mean extraction		Turns	
Water available	Gravity	Drip installed	Gravity	Drip installed
0–3	0.68	0.87	232	252
4–6	3.90	4.33	141	216
7–9	4.96	4.79	115	140
10–13	5.56	5.20	144	221
14 or more	6.36	5.53	605	565
Within-farmer installation coefficient: −0.278 (SE 0.153)				

Notes: Paid sessions, 2,631 feasible extraction turns; 14 infeasible turns excluded. Upper rows are unadjusted means. The regression includes farmer effects, available-water indicators (capped at the card endpoint), session and round effects, with standard errors clustered by game. Installed farmers retain the gravity option. The installation coefficient is an association.

6.2 Upstream installations and the water below

Water moves only downstream, so an installation can change the water of farmers below it but not above it. Table 7 tests this direction. Because the number of farmers above a seat rises mechanically toward the tail, every specification includes seat-by-session effects as well as session-by-round effects.

Table 7: Upstream installations and the water below: associations

A. A farmer's extraction, with farmer effects	Without water control	With water control
Installed farmers upstream	0.328 (0.186)	0.137 (0.084)
Installed farmers downstream	−0.321 (0.182)	−0.145 (0.079)
Own installation	−0.401 (0.237)	−0.331 (0.116)
B. Water reaching farmers below, per installed farmer upstream	Across groups	Within groups
Water reaching seats 2–5 (units)	0.832 (0.198)	0.138 (0.323)
Water reaching the tail (units)	0.447 (0.125)	−0.179 (0.169)
Tail receives 4 or more units (share)	0.100 (0.024)	−0.023 (0.036)

Notes: Coefficients, with game-clustered standard errors in parentheses; 38 games. Panel A: 2,631 feasible paid-session turns. It includes farmer, seat-by-session and session-by-round effects; the water control adds indicators for every observed available amount. Panel B: 2,116 farmer-rounds at seats 2–5 and 529 complete group-rounds for the tail. "Within groups" adds game effects.

The direction is as the canal predicts. A farmer extracts more when more farmers above them have drip and less when more farmers below do, by about a third of a unit per installed farmer in each case (panel A). The estimates are imprecise, with t-statistics near 1.8. They shrink when the water available to the farmer is held fixed, as they should if the channel runs through water passed down. Across groups, each installed farmer upstream goes with 0.8 more units reaching the farmers below and 0.45 more at the tail. The share of rounds in which the tail can use drip is also 10 percentage points higher for each installed farmer upstream (panel B). In rounds

with no installed farmer upstream the tail had usable water 40 percent of the time; with all four installed, 70 percent.

Within groups the association is imprecise (0.14 units, standard error 0.32). Groups that adopt also share differently, so the data cannot tell how much of the extra water comes from drip and how much from groups that adopt and share together. For the farmer below, either source can make an upstream installation a signal of more water to come. The signal is consistent with tail purchases following upstream purchases (Section 4). In the canal model, the farmers below use almost all of what is released (Section 6.3).

6.3 In the canal model, farmers below use almost all of a release

We use the canal model of Section 4. Its upstream rule is calibrated on both paid sessions of the same games, so its accounting is conditional on that rule. Its excess terms were estimated from actual play, where installed farmers could choose either card. For each seat, and at the empirical supply of each session, we compare an all-gravity canal with one installation at that seat. At upstream seats, only the adopter's mean excess term changes. Both paths use common random draws from the gravity residual distribution. The response to the excess already taken above, γ, passes the change down the canal. Supply is held fixed within each comparison. Appendix B adds a simple tail rule estimated relative to the card optimum and water reaching the tail.

Own saving is the reduction in the adopter's extraction. Downstream uptake is the sum of the increases below. Net group extraction is uptake minus saving. Game-cluster bootstraps recalibrate the canal model and resample supply.

At seats 1–3, own saving is 0.33–0.56 units and uptake below is 0.32–0.54 across the two sessions. Uptake absorbs 95–96 percent of the releases. Figure 7 shows the accounting.

Net group extraction falls by only 0.02–0.03 units. In all six comparisons the interval for the net change excludes zero. At seat 4 the release is smaller and its interval includes zero. A tail installation has no downstream uptake, and the change

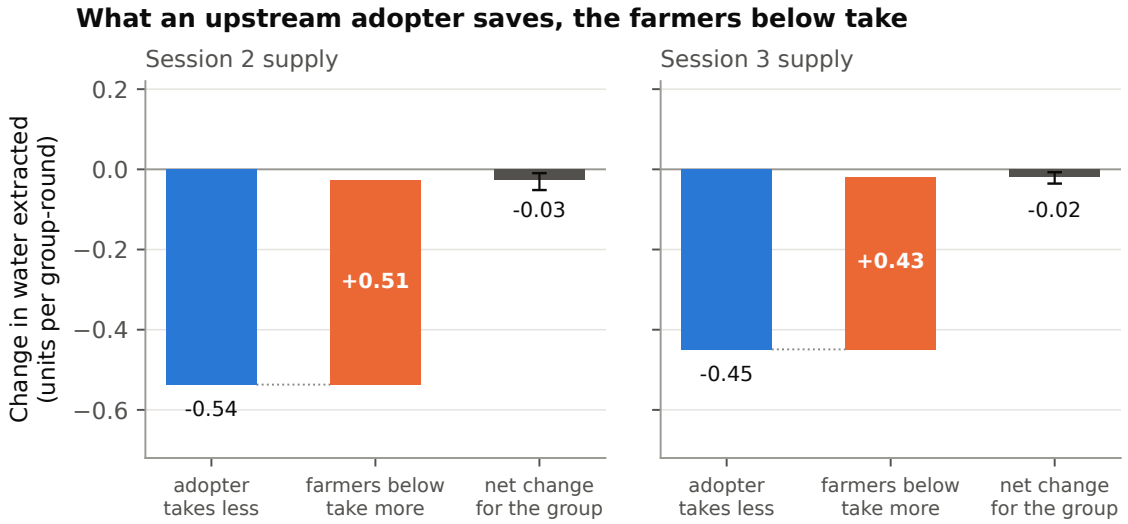


Figure 7: What an upstream adopter saves, the farmers below take. In the canal model at each paid session's empirical supply, one installation at seats 1–3 (averaged) is compared with an all-gravity canal. Bars give the adopter's reduction in extraction, the extra extraction by the farmers below, and the net change in group extraction with its 95 percent game-bootstrap interval, in water units per group-round.

in own extraction at the tail is imprecise. Appendix Table B3 reports all quantities and intervals, including small or negative values.

6.4 Who benefits when someone installs?

Table 8 relates installations at each seat to the inequality of extraction in each group-round, to the water delivered to the tail and to the gap in extraction between head and tail. The regressions include game and session-by-round effects. Installations at the head and seat 2 are associated with more equal extraction. Installations at seats 3 and 4 are associated with less water reaching the tail. At seats 3 and 4 water is moderate and drip makes each unit most valuable (Figure 2(b)), so an installed farmer there may take more, not less.

Table 8: Installation locations and the distribution of water: associations

Installed seat	Extraction Gini	Tail water	Head–tail gap
1	−0.052 (0.022)	0.069 (0.331)	−0.729 (0.551)
2	−0.054 (0.030)	0.646 (0.437)	−0.212 (0.549)
3	0.051 (0.028)	−0.605 (0.367)	0.265 (0.591)
4	0.023 (0.024)	−0.913 (0.403)	1.107 (0.653)
5	0.005 (0.032)	0.027 (0.477)	−0.559 (0.593)

Notes: Coefficients, with game-clustered standard errors in parentheses; 515 complete feasible paid group-rounds, 38 games. Each outcome is regressed on all five installed indicators, game effects and session-by-round effects. Tail water is water available before the tail extracts. The gap is head minus tail extraction. Gini is computed across the five extractions; it is set to zero if all are zero. Appendix Table B5 gives intervals and tail-extraction results.

The canal model isolates a different comparison. Installing at seats 1–3 raises water reaching the tail by 0.21–0.35 units and lowers the extraction Gini by about 0.005–0.007. A tail installation leaves upstream use and the water reaching the tail unchanged. Appendix Table B4 pairs the Gini with absolute tail water and extraction. More equal extraction is not itself a welfare measure, because production differs across cards and water levels.

The observed associations and the canal model agree only in part. Both single out installations at the head and seat 2 as the ones that leave water for others. The canal model also raises tail water for installations at seat 3, where the observed association is negative. At seat 4 the observed association is clearly negative. Two explanations are compatible with the data. A farmer at a moderate-water seat who buys drip may be one who intends to use more water. Drip also raises the value of an extra unit most at moderate availability (Figure 2(b)), which pulls an adopter toward taking more. The canal model captures the pull only on average. The modeled gain to the other four farmers from a seat-4 installation turns negative when supply is scarce (about -0.1 tokens per round at 18 units and -0.6 at 12).

The canal model contributes the direction in which water moves. The observed associations show that the effect of an installation on others depends on how constrained its user is. In the terms of Section 4, drip and sharing are complements at the plentiful head and substitutes where water is moderate.

6.5 Demand and group production gains pull apart

Adoption also decides who gains from the water it releases. Initial demand concentrates downstream, while the installations that change other farmers' water are upstream. We turn the water accounting into production gains by pricing each farmer's extraction in the canal model on the printed cards. For each seat, Figure 1(b) compares one installation with an all-gravity canal at Session 2 supply. It splits the change in production into the buyer's own gain and the gain to the other four farmers. Their sum is the gross group production gain of the installation. The measure excludes the costs of exceeding a fair share in equation (1) and the

installments, which are the same at every seat. It locates where an installation adds most to production, not whether the installation pays.

Upstream, the buyer's own gain is about 5 tokens per round, and the gain to the other four farmers is another 2.3–2.6, through the water the buyer leaves. The group production gain at seats 1–3 is 7.7–7.8 tokens, with intervals of about 7.0 to 8.7. At seat 4 less water is released, and the gain to the other four farmers falls to 1.0. A tail installation has no one below it, so its group production gain, 3.1 tokens, is the buyer's own gain. Observed purchases run the other way (panel b), from 29 percent at the head to 58 percent at the tail.

The comparison connects to the bet of Section 4. For a farmer expecting about ten rounds of play, buying requires a gain of roughly 6 tokens per round. An upstream buyer's own gain of about 5 tokens falls short of what buying requires. The group production gain of about 8 exceeds it. Upstream adoption is worthwhile for the group but marginal for the buyer, because a third of the group production gain goes to the other four farmers. At the tail, under the sharing typically observed, the buyer's own gain and the group production gain are both lowest. Tail farmers who buy are betting that upstream farmers will hold back more than usual.

Figure 1 evaluates a first installation in an all-gravity canal. At the first decision, however, 21 of the 22 tail farmers who bought had seen an upstream purchase. We therefore switch each farmer's installation on and off while holding the other four at their observed first-decision purchases. The ranking of seats is unchanged (Appendix Table B6). In Session 2 the group production gain is 8.0 tokens at seats 1–3, 7.4 at seat 4 and 3.8 at the tail. Upstream purchases raise the tail buyer's own gain from 3.1 to 4.0 tokens, but not to the level of an upstream buyer's.

The gap depends on water (panel c). With a constant supply of 30 units, an installation at the head adds 8.8 tokens to group production and one at the tail 3.6. With 12 units the gains shrink to 1.5 and 0.7, because the farmers below hold too little water for drip to make much difference. The upstream gain comes from the card logic of Section 4. With plentiful water, keeping to a fair share of the water left costs 8 tokens on gravity but 2 on drip, so drip makes sharing cheap. The farmers

below receive what the adopter no longer takes. At the scarce tail, drip raises the value of each unit but frees nothing for anyone.

Groups that adopt may also share differently in ways the canal model does not capture, as the estimates within groups in Table 7 suggest. The comparison therefore locates the gap between demand and group production gains. It does not measure the size of the gap in the field.

6.6 Productivity and conservation are different outcomes

Downstream uptake can make a release useful to another farmer while limiting the reduction in group extraction. The planner benchmark constructed in Section 4 raises output in a different way, through coordinated technology and sharing with the same water. Both results concern extraction, which is not the same as field evapotranspiration or basin depletion. Conservation programs must say where released water goes and must measure delivery, extraction and depletion separately from counts of installations.

7 Discussion: the combined evidence, the field, policy and limits

7.1 What the combined evidence explains

The results fit one account: the allocation of water along the canal decides who wants drip, whether it pays and what it achieves. Each result does a different job.

Position was randomized, and it doubles first-decision demand from head to tail, so fixed differences in soil or farm productivity cannot explain the gradient. At the tail, buying follows the order that the water bet predicts. One of 10 farmers bought without an upstream buyer, 11 of 38 head farmers bought, and 21 of 28 tail farmers bought after an upstream purchase. Repurchase followed the water a buyer actually received: 89 percent among downstream buyers with usable water in most of their drip rounds, against 55 percent otherwise. When supply fell in Session 3, demand peaked at seat 4, where the water incentive moved, and not at the last seat. In

the field, farmers at real canal tails are readier to accept drip and gain less from it (Section 7.2).

The same account carries over to water use. With 4 to 7 units, drip raises output without changing the withdrawal that maximizes private production. Farmers with plenty already hold back close to a fair share, so drip releases little. Farmers below use almost all of what is released. Land and crops are fixed in the game, so no expansion is needed for the efficiency paradox.

Personal and group behavior complete the account. The private benchmark predicts the direction of responses, not their full size. A persistent personal part and alignment within groups, stronger among farmers with social ties, explain much of who buys at a given seat. Randomly reordering farmers within groups recreates most of the pooled association between seeing an upstream buyer and buying. Alignment within groups therefore carries that association as much as the order of choices does.

Three questions remain open. Upstream farmers value drip closer to what they could take than to what they share, and the data explain this only in part. The tail's response to upstream buyers mixes information about water with alignment in the group. Experienced water was not assigned, because buyers chose to buy and stayed in their groups.

7.2 Canal position in the field

Real canal tails show the pattern of the game. The accompanying impact evaluation surveys the 914 farms from which the players were recruited [19]. Among farms that had never used drip, plots at a canal tail go with a 7.6 percentage point higher rate of accepting the first randomized subsidy offer, within districts and with farm controls ($p = 0.04$). Raw acceptance by the main plot's position is 33 percent at the tail and 31 percent at the head. Ever having used drip does not differ with tail position. Among cotton farms that have used drip, 38 percent of low performers but 8 percent of high performers lie at a canal tail ($p = 0.02$). The association between the 2024 cotton-drip share and yield is 0.16 log points lower at the tail ($p = 0.04$). Tail farmers are readier to take up drip and gain less from it, as tail farmers do in the game. In the game, linked farmers whose own plot lies at a canal tail also buy more

beyond the private benchmark (Section 5). The field comparisons are observational, and canal position may also capture the timing and quality of water.

7.3 Real drip users in the game

Economists disagree about whether people behave in experimental games as they do outside them. Behavior in a trust game predicted loan repayment among microfinance borrowers [16]. Laboratory measures of social preferences, however, did not carry over to market transactions [18] or to public-goods behavior in the field [31]. Real drip users give an individual comparison. We follow the impact evaluation in defining drip adoption as ever having used drip on the farm, the definition behind the field rates in Table 10 [19]. Seventeen of the 142 linked participants have used drip, and 11 still do. All 11 current users installed drip before the games, between 2019 and 2024, so playing the game cannot have caused their installations. Table 9 compares their behavior in the game with that of other linked farmers. We include seat-by-session and session-by-round effects, cluster standard errors by game, and report each comparison across all farmers, within region and within group.

Table 9: Farmers who use drip on their own farms: behavior in the game

	Across farmers	Within region	Within group
Currently use drip (11 farmers)			
Practice extraction at the same water (units)	−0.45 (0.26)	−0.45 (0.21)	−0.57 (0.30)
Paid extraction on gravity turns, same water (units)	−0.01 (0.16)	−0.05 (0.26)	−0.55 (0.21)
Maintenance investment, paid sessions (tokens)	1.21 (0.99)	1.20 (0.75)	0.70 (0.60)
Purchase at the first decision (probability)	0.22 (0.16)	0.21 (0.15)	0.06 (0.16)
Ever used drip (17 farmers)			
Practice extraction at the same water (units)	−0.32 (0.21)	−0.18 (0.21)	−0.48 (0.25)
Paid extraction on gravity turns, same water (units)	−0.12 (0.13)	−0.02 (0.19)	−0.48 (0.18)
Maintenance investment, paid sessions (tokens)	1.38 (0.72)	1.03 (0.56)	0.59 (0.51)
Purchase at the first decision (probability)	0.12 (0.12)	0.09 (0.13)	0.02 (0.13)

Notes: Coefficients on the drip-use indicator, with game-clustered standard errors in parentheses; 142 linked participants in 38 groups. Ever used drip is the impact evaluation's adoption measure (module D2); current use adds a current-use answer of yes. Turn-level regressions with seat-by-session and session-by-round effects; extraction rows add indicators for the water available. The practice session came before drip was available. Participants with unknown survey status are excluded, not coded as nonusers. All estimates are associations.

Farmers who currently use drip took about half a unit less water per turn in the practice session, with the same water available (3.6 units against 4.9 in raw means). The practice session came before drip existed in the game. Within groups they also took about half a unit less on gravity turns in the paid sessions. They invested somewhat more in maintenance and bought drip somewhat more often.

Both differences are imprecise. When we add the six farmers who stopped using drip, most of them in Khorazm, the differences point the same way but are smaller. The comparisons rest on few farmers. Their direction is consistent. Farmers with field experience of drip behave in the game as water conservers.

An earlier presentation of the same data reported larger differences. It identified drip from crop irrigation records and coded unlinked participants as nonusers. We now use the adoption module of the survey, exclude unlinked participants and account for the grouped design.

Section 5 relates field experience, real canal position, willingness to pay and beliefs to buying beyond the benchmark. They point the expected way but explain little of the personal part. Appendix E also repeats the comparisons of Table 9 for cluster membership, plans to adopt drip and refusal of subsidized drip.

7.4 Regional correspondence with the field

The game places all farmers in the same canal and presents the same technology cards. Outside the game, their experience differs. Table 10 puts the game outcomes beside field adoption in all four regions. Khorazm is coded Xorezm in the game data; we use Khorazm in the text. The field column measures ever having used drip, while the game columns measure purchase at the first decision of a session. Their levels are not directly comparable.

Table 10: Field adoption and game purchases in all four regions

	Fergana	Kashkadarya	Samarkand	Khorazm
Survey farms	232	227	229	226
Ever used drip in the field (%)	4.3	7.0	14.4	19.9
Game groups	10	11	8	9
Session 2 first purchase (%)	40	31	50	44
Session 3 first purchase (%)	44	42	42	67
Session 2 head buyers	3/10	1/11	2/8	5/9
Session 2 tail buyers	6/10	6/11	8/8	2/9
Session 3 repurchase (%)	73	63	75	93
Session 3 entry (%)	21	21	10	24

Notes: Field rates are regional aggregates from the 914-farm survey reported by Liutin et al. [19]; they are not calculated from the linked game subsample. Game rates are first-decision purchases. Repurchase conditions on any Session 2 purchase; entry conditions on no Session 2 purchase. Each region–seat cell has one farmer per group. Kashkadarya and Khorazm correspond to the panel's Quashqadarya and Xorezm labels. Comparisons are descriptive.

Khorazm has the highest field adoption, at 19.9 percent, and the highest rate of purchase at the first decision of Session 3, at 67 percent against 43 percent elsewhere.

In the field survey, however, its farmers have the least favorable expectations about the effect of drip on yields. According to the draft of the accompanying impact evaluation, the government in Khorazm has more influence on installation, and farmers have little choice of installer [19]. Farmers there also worry about salinity and about whether their crops suit drip. They may adopt drip because they are familiar with it and with the institutions around it, without being confident that it pays. Their field experience gives context for their readiness to buy in the game. It does not identify the cause.

In Session 2, Khorazm also differs in who buys: its head buys more than its tail, 5/9 against 2/9, while the tail buys most elsewhere. The regional version of the model in Section 4.6 reads the reversal as pessimism about upstream water sharing, even though water was relatively reliable in the Khorazm games.

The Khorazm farmers who played also differ from the other players. Among the Khorazm players linked to the field survey (39 of 45), the median farm has 46 hectares, against 30 elsewhere. On their main plot, 64 percent report medium or high salinity, against 34 percent. Only 19 percent plan to adopt drip, against 68 percent, and 69 percent refuse it at every subsidy level offered, against 16 percent. In the game, however, 91 percent of Khorazm players bought drip at least once, against 61 percent elsewhere. Their objections in the field (salinity and the effect of drip on yields) do not arise in the game, whose payoffs are printed and whose land is not saline. In the game they respond to its economics and to their group. What holds Khorazm farmers back in the field may be their land and not an aversion to drip. Only 24 of 55 Kashkadarya players are linked to the survey, and refusal in the survey does not predict whether the same farmer buys in the game.

The agreement between game and field across regions is limited. Most of the variation between games lies within regions, and in the field, adoption means a durable installation under local institutions. We read the regional pattern as a correspondence with field experience. We do not read it as validating the game as a forecast of individual adoption.

7.5 Technology and allocation must work together

The planner benchmark constructed in Section 4 combines drip with equal sharing and raises group production by 34 percent with the same water. The benchmark is conditional on full supply and coordinated sharing.

Part of the distance between the benchmark and actual purchases comes from allocation. At the head, a farmer who compares drip with the best feasible withdrawal on gravity sees an extra 5 tokens per round and a payback of nine rounds. At an equal share of 6 units, the extra output is 11 tokens and payback is about four rounds. The tail can obtain the larger gain only if all upstream farmers leave enough water. Each farmer can buy the equipment, but no farmer can buy everyone else's adherence to equal sharing. A subsidy for the technology alone does not secure equal sharing.

Private and group incentives also pull apart across seats (Section 6.5). Where water is plentiful, an upstream installation adds about 8 tokens per round to group production, a third of it for the farmers below. The buyer's own gain, however, falls short of what buying requires, and few upstream farmers buy. At the tail, where most farmers buy, an installation adds 3–4 tokens, all to the buyer. A subsidy that follows demand will therefore concentrate installations where, under typical sharing, they add least to group production.

The game and the field point to one paradox of adoption. Drip is wanted most where water is already short: at the tail of the game canal, and among never-users at real canal tails, who accept subsidized offers more readily. For a farmer short of water, drip raises output with little reduction in use, because the most profitable use is all the water received. It releases little water for others. The purchase is a bet on upstream sharing, and a failed bet is costly. In the game a failed tail bet loses the installments, and in the field drip goes with smaller yield gains on tail farms [19]. Severe shortage can also block adoption. In the same survey, farms that report a water-shortage shock in the last five years are 9.5 percentage points less likely to have used drip. Farmers with water to spare gain least from drip and buy least, although their adoption could release water. Falling supply can strengthen

their demand by bringing them into the range of water where drip adds most to private production. In California, droughts accelerated the diffusion of drip [29, 30]. A program that follows demand therefore funds drip first where it saves little water, and falling supply, as in a drought, can raise upstream demand.

A release can benefit only farmers farther downstream. Whether an upstream installation produces the largest group production gain depends on how farmers below respond and on the allocation. Under the equal-sharing benchmark, every installation adds the same 11 tokens. The results give a reason to consider and to test upstream installation combined with allocation rules that keep released water flowing past the next upstream withdrawal to the farmers below. They do not establish that such targeting would align private and group incentives in the field, where upstream farmers often have secure access.

Universal installation is also conditional on the supply. With 20 units, the output-maximizing allocation uses four drip systems and gives the fifth farmer no water. The benchmark raises a question of distribution alongside the efficiency calculation. When the planner chooses how many installations to use, nothing guarantees that every farm gets water. The lesson is to assess technology together with the water delivered and the rule that distributes it.

Programs should therefore distinguish at least two goals: raising agricultural output with the existing supply, and reducing water extraction or depletion. In the game, equal sharing with drip can raise output while using all the water. For the second goal, the released water must stay in the system, and farmers downstream must not extract it again. Allocation rules that specify delivery, withdrawal limits and the destination of released water complement support for equipment. Evaluations should count installations and measure crop output, and they should also measure the water delivered and used across the canal. The game does not select a legal or administrative instrument. It shows why a target for installations is an incomplete target for saving water [13, 11].

Allocation rules must distinguish an entitlement from a fair share of the water left. When one upstream farmer takes too much, farmers below who keep to a fair share of the water left can still pass most of the shortfall to the tail. Predictable

delivery would make drip a better investment at the tail. Conservation also requires the rules to specify whether a release remains in the system or is withdrawn below.

Visible early adopters motivate a separate field test. The test should assign exposure to demonstrators and water delivery independently, and record purchases and water use. The test would show whether a demonstration changes take-up, improves allocation or reveals a group already willing to buy.

7.6 Scope

Table 11 sets out how strong the evidence is for each main claim. Results that rest on voluntary adoption, purchase history or group behavior are associations. The experiment randomizes position, not peers, so the clustering of purchases is an association whose mechanism we do not separate (discussion between decisions, shared local conditions, or responses to visible choices). All groups played the sessions in the same order. Experience, longer play and a falling water cap were not assigned separately. The model exercises reuse the same games and are exploratory. We did not ask farmers how long they expected play to last or how they expected upstream farmers to share water. The model fits both beliefs to first-decision purchases. Inference rests on 38 groups recruited by availability. With so few groups, we use randomization inference and group-level resampling. Appendix A documents how we reconstructed the protocol, including manual stops and incomplete dates. Most linked survey interviews preceded the games, but not all. Social ties come from a questionnaire after the games, and we chose the tie categories after seeing the data. We also chose the comparison of clustering at the head and the tail after seeing the data, and several survey comparisons rest on few farmers.

Randomized position raises demand for drip toward the tail, where purchases are bets on upstream water sharing. Beyond position, the largest part of who buys stays with the individual farmer, part is shared within groups, and cooperation in sharing water does not carry into buying. The group production gain from adoption runs the other way. It is highest upstream, where water is plentiful and drip makes sharing cheap, and lowest at the tail, where an installation frees water for no one. Farmers below use the released water, so gains in productivity come with only a

Table 11: Claims and strength of evidence

Claim	Evidence and limit
Position effect	Randomized seat assignment; strongest causal result. Position combines water, turn order and observed choices.
Repurchase	Large descriptive history split; prior purchase was voluntary.
Clustering	Strong agreement within groups; mechanism unresolved.
Personal and group parts of buying	Variance shares from pairs of decisions; the personal part mixes stable beliefs with each farmer's own experience.
Reliable water and repurchase	Suggestive association in selected downstream buyers.
Own-extraction association	Small within-farmer reduction (0.28 units); installation chosen.
Upstream adoption and water below	Strong across groups (0.8 units per installer); imprecise within groups.
Real drip users in the game	Less water, clearest for 11 current users; maintenance and purchases imprecise; association.
Downstream re-extraction	Conditional canal-model result; 95–96 percent of a release at seats 1–3 is used below.
Demand vs group production gain by seat	Demand gradient randomized; group production gain from the calibrated canal model, robust to observed configurations.
Regional beliefs	Exploratory as-if fit to four small regional samples.

small reduction in extraction. Farmers can buy the technology, but not the allocation that decides whether it helps them or anyone else.

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A Protocol, cards and payment

A.1 Session sequence and information

The same five farmers remained together, with seats randomly reassigned between sessions. Each round proceeded from seat 1 to seat 5, with investment, any eligible purchase, and extraction completed before the next farmer acted. All decisions were visible. Talk was allowed between decisions and between sessions, but not during a decision. Production cards were printed and explained. Neither the endpoint nor a stopping probability was supplied to participants.

Table A1: Session sequence and realized duration

	Session 1	Session 2	Session 3
Purpose	Practice	Drip introduced	Drip offered anew
Performance payment eligible	No	Yes	Yes
Technology menu	Gravity	Gravity and drip	Gravity and drip
Water cap in round t	30	30	$\max\{0, 30 - 2(t-1)\}$
Mean realized rounds	3.76	5.55	8.37

Notes: 38 groups in each session. The means are realized durations, not announced horizons. Drip expires at each session boundary. Every farmer starts each paid session without an installation.

A.2 Infrastructure and printed output

Infrastructure starts each session at $H_{s1} = 75$ on a scale from 0 to 100. With individual investment $I_{ist} \in [0, 20]$, the maintained transition and delivered supply are

$$H_{s,t+1} = \min\left\{100, \max\left[0, H_{st} - 25 + \tfrac{1}{3}\sum_{i=1}^{5} I_{ist}\right]\right\}, \tag{3}$$

$$W_{st} = \min\left\{\frac{40}{1 + \exp[-0.2(H_{st} - 60)]}, \bar{W}_{st}\right\}. \tag{4}$$

Investment of 75 tokens in total offsets depreciation of 25 infrastructure points. The session cap $\bar{W}_{st}$ is given in Table A1. Current investment affects the next round, whereas extraction immediately reduces residual water. These equations describe the implemented supply rule; they are not measured subjective forecasts.

Table A2: Exact printed integer production cards

Water used	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Gravity	0	0	8	15	22	27	32	35	38	39	40	39	38	35
Drip	0	0	0	13	27	37	43	45	44	43	40	37	32	27

Notes: Entries are production tokens before investment and installments. An installed farmer may use either card. Figure 2(b) compares best feasible extraction at the water available, rather than equal extraction on the two cards.

A.3 Token account and cash payment

For endowment 20, investment I and chosen card output $P(c)$, the round account is $\max\{0, 20 - I + P(c) - f\}$, where $f = 15$ in an installation's first three rounds and zero afterward. Installation does not constrain the current investment endowment. One round drawn from the combined realized rounds of Sessions 2 and 3 supplies performance pay, at 4,000 UZS per token. The attendance payment is 200,000 UZS. The maximum round account is 65 tokens, so the performance component is at most 260,000 UZS and total compensation ranges from 200,000 to 460,000 UZS. These protocol units do not turn a sum of session tokens into a cash willingness to pay.

The protocol specifies chance determination of the total rounds and a 30 percent stopping probability (study protocol, paragraph 23). The design set minimum lengths of 3, 5 and 6 rounds for the practice session, Session 2 and Session 3. After the minimum, the software drew a stop after each round with a settable probability, default 0.3. Five sessions ended before their minimum by procedural error: three Session 2 games at round 4, and one Session 3 game each at rounds 4 and 5. Facilitators also stopped sessions by hand. Practice sometimes ended accidentally or when moving on, and many third sessions ended around round ten to preserve questionnaire time, which was not a design feature. The realized lengths cannot identify the configured probabilities. Farmers knew neither the endpoint nor the stopping rule and were told after each session only that the next would be longer.

Installments still due when a session ended were not charged. All hypothetical first-decision purchases in Table 1 have at least three played rounds; five actual late purchase spells have fewer than three installed rounds. The round account applies its zero floor separately to each alternative. Enumerating allowed water, investment and installment states leaves the paid-round purchase range at -15 to $+11$ tokens. The game-master ledgers record -15 in exactly the first three installed rounds, consistent

with the author's confirmation of 3×15; the 14-upfront or 6×3 price and ten fixed rounds in an initial memo and presentation were abandoned.

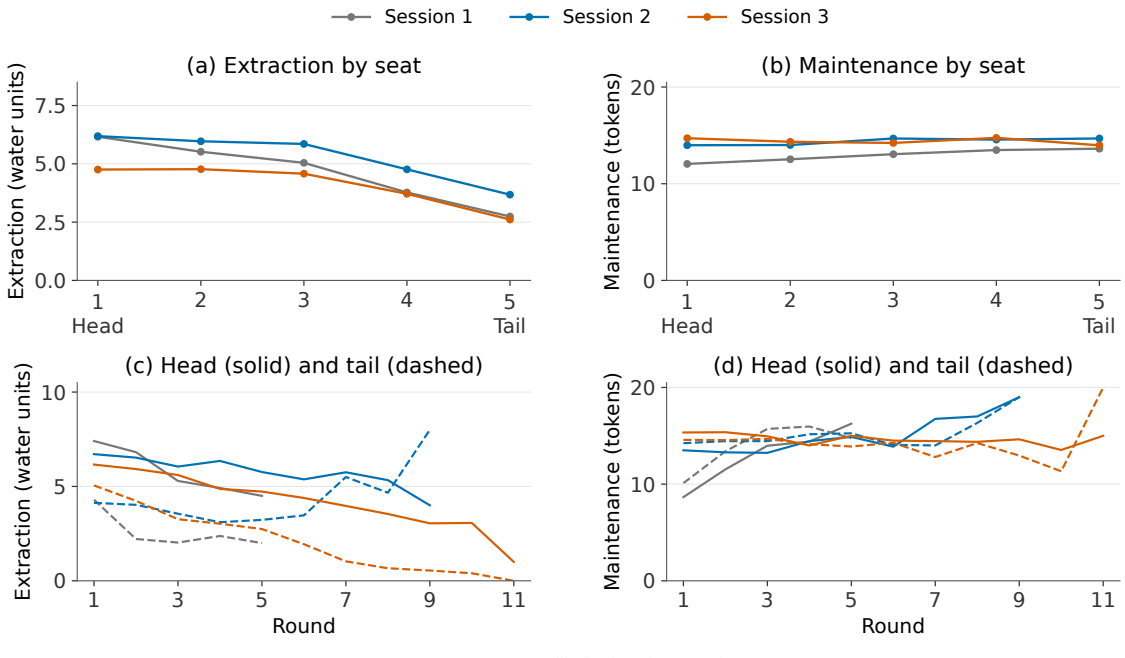
A.4 Dates and survey chronology

Groups were recruited by availability. Refusals and unsuccessful contacts were not recorded systematically, so the study has no response rate.

Timestamped exports matched to group-labelled workbooks, supplemented by consistent workbook creation properties, date 14 of 38 groups to 4–20 December 2024. This assumes a correct controller clock and contemporaneous saving. The dated groups comprise 5 of 10 in Fergana (18–20 December), 2 of 11 in Kashkadarya (16–17 December), and 7 of 9 in Khorazm (4–12 December). No file date was recovered for the eight Samarkand groups; according to the field team, they were played in the middle of the campaign, between the Khorazm games and those in Kashkadarya and Fergana. The remaining groups have no recorded date, and no analysis in the paper depends on exact play dates.

Recorded starts for all 914 survey records span 30 October 2024–21 April 2025; the 142 linked participants span 7 November 2024–12 April 2025. Of the 60 linked participants with both an interview start and an inferred game date, 39 starts precede play, 12 follow it and 9 are on the same day. The other 82 linked participants lack a recovered play date. Using interview ends gives 38 before, 12 after and 10 on the same day. These timestamps do not establish the chronology of the original interviews; they rule out describing the linked survey as uniformly pregame. The exact node crosswalk gives 142 participants in 139 households, with 48 links unknown. Date evidence and regional aggregates are documented in the accompanying date audit.

B Extraction and water accounting



Groups still playing, by round

	1	2	3	4	5	6	7	8	9	10	11
S1	38	38	38	25	4	0	0	0	0	0	0
S2	38	38	38	38	35	16	4	3	1	0	0
S3	38	38	38	38	37	36	31	24	22	15	1

Figure B1: Extraction and maintenance by canal position. Upper panels show session means by seat. Lower panels show round means at the head and tail; numbers below give groups still playing in Sessions 1/2/3. Extraction uses feasible turns; maintenance uses all recorded turns. Late rounds rest on few groups.

B.1 One canal rule and the tail extension

The upstream rule is shared with Section 4. For seat $k < 5$, residual water A_k, fair share $q_k = A_k/(6-k)$ and cumulative excess $X_k = \sum_{j<k}(c_j - W/5)$,

$$c_k = \text{clip}\{q_k + \alpha_{d_k,b(A_k)} + \gamma X_k + \varepsilon_k, 0, \min(A_k, 13)\}.$$

Water passes sequentially as $A_{k+1} = A_k - c_k$. Ordinary least squares on complete paid rounds with feasible upstream actions estimates the technology–water-band excesses and γ. They describe actual installed farmers' full operating menu, including gravity. This is a reduced-form representation of holding back as in equation (1), not an estimate of its preference parameter.

The upstream rule ends before the tail. We estimate a simple tail extension from 515 feasible tail choices in those rounds. Let $c_d^*(A)$ be the feasible production

Table B1: Upstream excess over a fair share of water left

Water available	0–6	7–9	10–13	14 or more
Gravity	−0.17	0.71	0.73	0.86
Drip installed	0.06	0.41	0.40	0.25

Notes: 2,088 upstream decisions in 522 complete paid rounds; $\gamma = 0.271$. Cell coefficients are associations, not installation effects. Water bands use the cutoffs 6.5, 9.5 and 13.5 in simulation.

optimum with water A, retaining gravity when installed. For integer water it is $\min(A, 10)$ on gravity, $\min(A, 7)$ when installed with at least 4 units, and the gravity optimum below that. The largest feasible zero-output tie is used at one unit. We estimate ρ_d as the mean of $c/c_d^*(A)$ over positive optima, then simulate

$$c_5 = \text{clip}\{c_d^*(A_5)(\rho_d + \eta), 0, \min(A_5, 13)\}.$$

The residual $\eta = c/c_d^*(A) - \rho_d$ is pooled across installation states. It allows departures from the card optimum; clipping enforces water balance. The continuous simulation switches the fallback at 3.5 units, consistent with the integer-water convention. Zero-water observations contribute no ratio and imply zero extraction.

Table B2: Tail-rule calibration from paid-session play

	Turns	Positive optima	Mean water	Mean utilization ρ_d
Gravity	201	145	3.18	0.932
Drip installed	314	246	3.41	0.921

B.2 Comparisons and uncertainty

For each session we draw supply from its empirical distribution over the 522 eligible group-rounds. All five farmers start on gravity. Each comparison installs at one seat and resets to that baseline. At upstream seats only the adopter's α changes; the residual distribution remains gravity in both paths, using common uniform ranks within each realized water band. Downstream water bands and the γX response adjust as water passes down. At the tail only the menu optimum and fitted mean utilization switch, with the same pooled proportional shock. Thus uncertainty does not masquerade as a change in mean behavior.

Point comparisons use 131,072 common draws. Percentile 95 percent intervals use 999 resamples of 38 games; each resample refits upstream and tail coefficients and

residual pools and resamples the session supply distribution, with 8,192 common draws. The bootstrap uses identical simulation uniforms across draws to limit Monte Carlo noise. It describes sampling uncertainty conditional on this rule, including its tail extension, and not model uncertainty or a causal adoption response. The accounting identity is $\Delta C_{\text{group}} = \text{uptake below} - \text{own saving}$.

Table B3: One-adopter water accounting at empirical session supply

Session	Seat	Own saving [95% interval]	Uptake below [95% interval]	Net group [95% interval]
2	1	0.563 [0.224, 0.893]	0.536 [0.207, 0.856]	-0.027 [-0.054, -0.010]
2	2	0.548 [0.216, 0.868]	0.522 [0.203, 0.829]	-0.026 [-0.051, -0.009]
2	3	0.500 [0.197, 0.778]	0.474 [0.181, 0.735]	-0.026 [-0.050, -0.009]
2	4	0.216 [-0.022, 0.418]	0.198 [-0.021, 0.389]	-0.017 [-0.036, 0.001]
2	5	0.066 [-0.107, 0.196]	0.000 [0.000, 0.000]	-0.066 [-0.196, 0.107]
3	1	0.533 [0.213, 0.827]	0.512 [0.198, 0.794]	-0.021 [-0.041, -0.008]
3	2	0.482 [0.193, 0.734]	0.462 [0.182, 0.707]	-0.019 [-0.037, -0.008]
3	3	0.332 [0.121, 0.529]	0.317 [0.114, 0.504]	-0.016 [-0.029, -0.006]
3	4	0.077 [-0.112, 0.273]	0.071 [-0.105, 0.252]	-0.006 [-0.023, 0.010]
3	5	0.031 [-0.094, 0.119]	0.000 [0.000, 0.000]	-0.031 [-0.119, 0.094]

Notes: Water units per group-round. Each row is a separate comparison. Rounding can obscure the identity. Small and negative savings remain in the intervals. The tail has no downstream recipient.

Table B4: Conditional distributional changes when one farmer installs

Session	Seat	Tail water [95% interval]	Tail extraction [95% interval]	Gini [95% interval]
2	1	0.348 [0.148, 0.500]	0.321 [0.136, 0.454]	-0.007 [-0.012, -0.003]
2	2	0.334 [0.142, 0.484]	0.308 [0.127, 0.438]	-0.007 [-0.012, -0.002]
2	3	0.337 [0.141, 0.484]	0.311 [0.128, 0.439]	-0.006 [-0.011, -0.002]
2	4	0.216 [-0.022, 0.418]	0.198 [-0.021, 0.389]	-0.003 [-0.007, 0.000]
2	5	0.000 [0.000, 0.000]	-0.066 [-0.196, 0.107]	-0.001 [-0.004, 0.002]
3	1	0.283 [0.121, 0.399]	0.262 [0.110, 0.362]	-0.007 [-0.012, -0.003]
3	2	0.257 [0.111, 0.353]	0.238 [0.102, 0.323]	-0.007 [-0.011, -0.002]
3	3	0.208 [0.085, 0.300]	0.193 [0.078, 0.278]	-0.005 [-0.009, -0.001]
3	4	0.077 [-0.112, 0.273]	0.071 [-0.105, 0.252]	-0.002 [-0.005, 0.001]
3	5	0.000 [0.000, 0.000]	-0.031 [-0.119, 0.094]	-0.000 [-0.002, 0.002]

Notes: Same model draws and game bootstrap as Table B3. Tail water is available before tail extraction. Water changes are units per group-round; Gini changes are index points. An all-zero extraction vector has Gini zero. Equality is not a welfare measure.

Table B5: Distributional associations: complete intervals

Seat	Gini	Tail water	Tail extraction	Head–tail gap
1	−0.05	0.07	0.30	−0.73
	[−0.10, −0.01]	[−0.60, 0.74]	[−0.37, 0.97]	[−1.85, 0.39]
2	−0.05	0.65	0.61	−0.21
	[−0.12, 0.01]	[−0.24, 1.53]	[−0.27, 1.49]	[−1.32, 0.90]
3	0.05	−0.61	−0.43	0.26
	[−0.01, 0.11]	[−1.35, 0.14]	[−1.19, 0.33]	[−0.93, 1.46]
4	0.02	−0.91	−0.66	1.11
	[−0.03, 0.07]	[−1.73, −0.10]	[−1.44, 0.13]	[−0.22, 2.43]
5	0.00	0.03	−0.09	−0.56
	[−0.06, 0.07]	[−0.94, 0.99]	[−1.04, 0.86]	[−1.76, 0.64]

Notes: Point estimates and 95 percent intervals from game-clustered standard errors with a t_{37} reference. Same 515 complete feasible paid group-rounds, 38 games and fixed effects as Table 8. Voluntary-adoption associations; intervals are pointwise across the reported outcomes.

B.3 Production gains at observed configurations

Figure 1 evaluates one installation in an all-gravity canal. Table B6 instead toggles each farmer's installation while the other four keep their observed first-decision purchases, group by group, at each session's empirical supply. The canal model, pricing and draws are those of Figure 1. In Session 2 the tail buyer's own gain is 3.1 tokens in the 10 groups with no upstream purchase and 4.0 in the 28 with one.

Table B6: Production gains from toggling one installation, other farmers at observed purchases

Session	Seat	Buyer's own gain	Gain to the other four	Group gain
2	1	5.19 [3.82, 6.71]	2.83 [1.33, 4.58]	8.03 [7.24, 9.11]
2	2	5.27 [3.97, 6.74]	2.73 [1.30, 4.51]	8.01 [7.20, 9.05]
2	3	5.41 [4.14, 6.74]	2.56 [1.19, 4.20]	7.97 [7.21, 9.00]
2	4	6.15 [4.81, 7.58]	1.22 [−0.19, 2.55]	7.37 [6.60, 8.14]
2	5	3.76 [2.58, 4.96]	0	3.76 [2.58, 4.96]
3	1	2.82 [0.93, 4.55]	3.03 [1.49, 4.75]	5.85 [5.20, 6.54]
3	2	3.09 [1.45, 4.47]	2.75 [1.31, 4.33]	5.84 [5.13, 6.58]
3	3	3.68 [2.31, 4.86]	1.98 [0.80, 3.32]	5.66 [5.11, 6.29]
3	4	4.18 [2.83, 5.36]	0.63 [−0.50, 1.77]	4.81 [4.23, 5.56]
3	5	2.55 [1.58, 3.47]	0	2.55 [1.58, 3.47]

Notes: Tokens of production per round, averaged over the 38 groups' observed first-decision configurations, with 95 percent intervals from 199 game resamples that refit the canal rule. Production is priced on the printed cards; an installed farmer uses the better card at its extraction. Installments and the costs of exceeding a fair share are excluded.

C Economic accounts tested

Table C1: Economic accounts and their scope

Account	What it establishes
Horizon	Under the paid-round rule, longer expected play can make purchase pay; duration alone does not explain the position and repurchase patterns.
Cooperation counterexample	Reliable moderate water can make immediate purchase preferable to optimal waiting under the same payment rule. This is a constructed expectation, not an estimated belief.
Regional beliefs	Common choice noise with region-specific beliefs about continuation and upstream water sharing describes the regional tail contrast. Beliefs are inferred as-if quantities.
Learning and persistence	Reliable water is associated with repurchase. Public production cards leave allocation learning, expectations and stable willingness to buy unresolved.

These exploratory exercises reuse the same games; none is a new confirmatory sample. Exercises that required installed farmers to operate drip exclusively are not used in the current economic or water accounting. The retained calculations allow gravity at every installed turn and value purchases by the paid round.

C.1 Horizon, cooperation and persistence

Longer expected play can make purchase pay, while reliable moderate water can make immediate purchase preferable to waiting. A constructed example uses a continuation belief of 0.9 in both paid sessions, current tail water of 5 units, future water of 6, and investment of 15 tokens. With one round drawn from the combined sessions, buying immediately adds 2.401 tokens to the expected paid account, against 2.158 from optimal waiting, a difference of 0.243. Unpaid installments lapse at the endpoint, the gravity fallback is retained and the floor does not bind in this example. Waiting is solved backward over the continuing decision problem with a negligible terminal probability. This is a possible expectation, not an estimated farmer belief.

The expected per-round gain in equation (5) is generated from the pooled canal rule at calibrated behavior, without regional belief shifts. It is not a buy-minus-wait value. The history and visibility coefficients describe voluntary choices. Neither that equation nor the regional fit identifies whether reliable water changes expectations, stable tastes or social responses; measuring expectations before purchase would help distinguish them.

C.2 Regional beliefs under the paid-round rule

Table C2 reports the specification of Section 4.6. Every farmer values a first-decision purchase by equation (2), with geometric beliefs about continuation and no known minimum length. At a Session 2 decision both session lengths are drawn from the belief; at a Session 3 decision the realized Session 2 length is used. The expected gain comes from the canal rule calibrated on Session 2, with the regional shift added to every upstream farmer's excess. The purchase probability is $\Lambda(\lambda\,\Delta V)$. The common scale is $\lambda = 2.47$; the log-likelihood is -114.1 for 190 decisions and nine parameters. Intervals come from 200 game-cluster bootstrap draws within region. This is an exploratory as-if fit to purchases.

Beliefs about the horizon are similar across regions: 10–12 expected rounds, longer than sessions actually lasted in any region (5.4–5.9 rounds). At these beliefs drip must add 5.2–6.1 tokens per round to pay, just above the 5 tokens it adds at the head. That is why upstream farmers are close to indifferent and about a third of them buy. These requirements are close to the empirical benchmark at the start of Session 3 (5.6 tokens), as if farmers expected Session 2 to last about as long as Session 3 eventually did. They had been told only that it would be longer than the practice session.

Beliefs about upstream water sharing differ by region. Without upstream adopters, Samarkand's farmers act as if the tail would receive 5.7 units on average, and Khorazm's as if it would receive 2.8. Fergana (4.3) and Kashkadarya (4.8) fall in between. Game-resampled 90 percent intervals for the beliefs of Samarkand and Khorazm do not overlap. Khorazm's low tail rate is not a general reluctance to buy: its three upstream seats bought more often than those elsewhere, 14 of 27 against 24 of 87. The as-if beliefs run opposite to what each region's games delivered. Samarkand's upstream farmers exceeded a fair share by the most, 1.3 units on average, and its tail received the least water, 3.1 units. Khorazm's took the least, 0.2 units above a fair share, and its tail received the most, 5.2 units.

In Session 3, Khorazm's tail purchases rose from 2 of 9 to 6 of 9, while they fell in every other region: from 6 to 4 of 10 in Fergana, 6 to 3 of 11 in Kashkadarya and

8 to 5 of 8 in Samarkand. Seats were reshuffled, so the Session 3 tail consists mostly of different farmers, many of them repeat buyers. The change therefore does not show that the same farmers revised their beliefs. It shows only that tail purchases moved in the direction of each region's experience.

Valuing purchases by the paid round, with no time preference, fits as well as a version that discounts token sums at 0.95 (log-likelihood −114.1 against −114.3). With four regions and 8–11 farmers per region and seat, we read the results ordinally. Common preferences with regional beliefs about upstream water sharing reproduce the main regional contrast at the tail.

Table C2: Regional as-if beliefs (Session 2 fit) and what each region's games delivered

	Fergana	Kashkadarya	Samarkand	Khorazm
Continuation probability	0.91	0.90	0.91	0.92
90% interval	0.90–0.92	0.88–0.91	0.91–0.92	0.90–0.93
Expected rounds	10.9	10.0	11.6	12.1
Required gain (tokens per round)	5.7	6.1	5.4	5.2
Upstream excess shift (units)	−0.29	−0.46	−0.75	+0.27
90% interval	−0.80 to −0.08	−0.61 to −0.26	−1.18 to −0.51	−0.08 to +0.71
Expected tail water, no upstream drip	4.3	4.8	5.7	2.8
Session 2 as played				
Upstream excess over a fair share	1.04	0.55	1.26	0.20
Tail water	3.7	3.9	3.1	5.2
Session length (rounds)	5.4	5.4	5.6	5.9

Notes: Beliefs are inferred from first-decision purchases, not elicited. A negative shift means upstream farmers are expected to take less than the calibrated rule. Required gain is the per-round drip gain at which $\Delta V = 0$ for a Session 2 purchase. The excess over a fair share is extraction minus the water reaching the farmer divided by the number of farmers from that seat down, averaged over upstream decisions.

D Position, persistence and group comparisons

D.1 Randomization inference for position

The Session 2 first-decision seat slope is 0.07105 in purchase probability per seat. Under the null, hold each group's number of buyers fixed and place them uniformly among its five seats. The statistic is an integer weighted sum of purchases with centered seat weights $(-2, -1, 0, 1, 2)$. Convolving the within-group distributions gives an exact two-sided probability of 0.002005, including outcomes tied with the observed statistic. Section 3 reports this exact result.

D.2 Entry, repurchase and agreement within groups

Table D1: First-decision purchases and Session 3 purchase history

Seat	1	2	3	4	5
Session 2 purchase	11/38	12/38	15/38	17/38	22/38
Session 3, all	18/38	16/38	19/38	21/38	18/38
Session 3 entry	5/19	3/19	3/15	5/20	2/20
Session 3 repurchase	13/19	13/19	16/23	16/18	16/18

Notes: A previous buyer bought at any time in Session 2. Current seats were assigned anew, so previous-buyer denominators differ across current seats. Entry and repurchase together give the pooled Session 3 total.

Table D2: Group agreement at the first decision in Session 2

Buyers in a five-farmer group	0	1	2	3	4	5
Observed groups	9	3	11	10	3	2
Independent-binomial expectation	2.83	9.63	13.13	8.95	3.05	0.42

Notes: Binomial expectation uses the pooled purchase rate, 77/190. For inference, random allocation holds that total fixed; it implies 3.109 unanimous groups, compared with 11 observed. The exact upper-tail probability is 0.0000558. A fresh 100,000-draw permutation has 9 outcomes at least as extreme. The binomial expectation and the fixed-total permutation differ slightly because the latter conditions on the number of buyers.

D.3 Visibility and the group-effects reference

Table D3 reports an exploratory turn-order exercise. Each farmer retains their choice and completed purchase history, but the sequence within each group and session is reordered. Upstream exposure is then reconstructed. The comparison is descriptive: water and actual position can change choices, so these reorderings are not exact randomization inference for visibility.

Table D3: Upstream-buyer association under actual and random order

First-decision linear probability specification	Observed	Random mean	Random 95% range
Seat and session effects	0.147	0.143	[0.130, 0.157]
Seat and game–session effects	−0.271	−0.304	[−0.335, −0.270]
Seat, session and game effects	−0.037	−0.056	[−0.083, −0.031]

Notes: 380 first decisions, 2,000 reorderings. The fractions of reordered coefficients at least as large as observed are 0.239, 0.028 and 0.071, respectively. The ranges describe random orders, not confidence intervals for a causal response. Group effects create a negative reference by excluding own choice from the exposure while using it in the group mean.

D.4 Compact purchase description and predictive comparison

Let R_{is} mark Session 3 decisions by a previous Session 2 buyer, B_{is} count upstream first-decision buyers already seen, and e_{is} be the expected per-round production gain from drip at the decision. We simulate e_{is} with the pooled canal calibration of Section 4, calibrated behavior, the farmer's seat, the upstream installed configuration visible before acting, and that session's empirical supply distribution. Gravity remains available. A normal-group-intercept logit gives

$$\operatorname{logit}\Pr(D_{is}=1) = a_s + 1.86\frac{e_{is}}{10} + 0.70B_{is} + 2.81R_{is} + \boldsymbol{x}_{is}'\boldsymbol{b} + u_g, \tag{5}$$

where $\boldsymbol{x}_{is}$ contains position and a Khorazm indicator, and u_g is a normal group intercept. The gain is a production index, not a paid-round value or a measured belief. Its coefficient is 1.86 per ten tokens, with a game-bootstrap 95 percent interval from -1.22 to 5.31. Purchase history and upstream purchases remain associated with buying; chosen installation and exposure do not identify their mechanisms. The value measures tested do not explain the full purchase pattern.

Table D4 reports every coefficient. Inference refits the purchase equation in whole-game bootstrap samples, conditional on the pooled gain calibration. The household-component predictive comparison below uses a buy-minus-wait index and does not validate prediction with e_{is}. It asks whether the compact description predicts held-out groups as well as two generic benchmarks. The nine-feature benchmark is a logit on seat, session, water, visible purchases and purchase history, with normally distributed farmer and group intercepts. The fifty-feature benchmark is a pooled logit with a longer list of history terms. Each model is fitted without one of 36 blocks

of games, where games that shared a household form one block, and scored on that block by mean log loss.

Table D4: Compact purchase description with expected per-round gain

Term	Estimate	95% interval
Intercept	−1.770	[−3.758, −0.016]
Session 3	−1.296	[−1.938, −0.587]
Expected drip gain e, per 10 tokens	1.855	[−1.220, 5.310]
Upstream buyers already seen	0.699	[0.094, 1.094]
Seat minus one	−0.037	[−0.265, 0.255]
Own Session 2 purchase, in Session 3	2.811	[2.099, 3.704]
Khorazm	0.233	[−0.381, 1.142]
Group standard deviation	0.002	[0.002, 0.865]

Notes: 380 first decisions, 38 games. Normal-group-intercept logit with 41-node quadrature, as in equation (5). Intervals use 999 converged whole-game bootstrap refits, conditional on the generated gain calibration; Appendix B describes that calibration. The boundary standard deviation is not evidence of zero group heterogeneity. These estimates describe full-sample associations.

Table D5: Predictive comparison using a buy-minus-wait index

Model	All	Session 2	Session 3
Compact, including group term	0.5777	0.6531	0.5023
Nine-feature Gaussian	0.5786	0.6450	0.5121
Fifty-feature history logistic	0.5808	0.6563	0.5053

Notes: Exploratory comparison with a buy-minus-wait value under its specified operating assumptions. The compact models use this index, rather than the per-round canal gain e in equation (5). The nine- and fifty-feature models provide the corresponding benchmarks. Mean log loss is reported for the same 380 decisions in 36 household-connected holdouts. This comparison concerns that index; it does not assess prediction with the canal gain. Paired intervals for the full compact model against the nine- and fifty-feature models include zero. This is a predictive comparison under the specified models.

In the separate visibility exercise, the fixed comparison of buyer and decliner counts against position and water gives a log-loss difference of −0.0334, with interval [−0.0700, 0.0027]. Its upper endpoint crosses zero, so that test did not establish an improvement. In the first round, decliners seen equal seat minus one minus buyers seen; the decline coefficient cannot be separated from a position effect without additional restrictions.

D.5 Personal and group parts of purchases

The outcome is purchase at the first decision of Sessions 2 and 3: 380 decisions by 190 farmers. Groups keep the same five farmers in both sessions, and seats are reassigned at random. The private benchmark is the paid-round value of equation (2). Its expected drip gain is simulated from the Session 2 canal rule for each seat and each configuration of visible upstream purchases. Beliefs are common to all farmers: one

continuation probability, one shift in upstream extraction and one choice-noise scale, fitted separately to each session. In Session 2 the fit implies 10.3 expected rounds and a shift of -0.44 units per upstream farmer. In Session 3 it implies 11.5 rounds and -0.25. For Session 3 the length of Session 2 is set to its mean for everyone, so the benchmark varies within a seat only with visible upstream purchases.

The common part is the seat-by-session purchase rate plus the deviation of the benchmark from its seat-by-session mean. The benchmark enters with a coefficient of one, so imitation cannot load onto it. Residuals are demeaned by region, and the variance of the regional means is the regional part. Let C be the mean product of two residual decisions. The personal part is C for the same farmer in the two sessions, minus C for different farmers of the same group in different sessions. The persistent group part is that second term minus C for farmers of different groups in the same region. The within-session group part is C for different farmers of the same group in the same session, minus C for such farmers across sessions. Each part is divided by the variance of purchase.

Standard errors come from a jackknife that leaves out one group at a time. The personal part has a permutation $p < 0.001$ when Session 3 decisions are shuffled among the members of each group (2,000 draws). The two group parts together have $p < 0.001$ when farmers are regrouped at random within region with both of their decisions (2,000 draws; mean under regrouping 0.003).

Table D6 relates each farmer's buying beyond the benchmark, the mean residual over the two first decisions, to field experience, real canal position, stated willingness to pay, beliefs about drip and farm characteristics. Each row uses every linked farmer, with no controls, within region and within group. Farmers off canals and farmers who sat at the tail in practice keep indicators and are not dropped. The outcome is not demeaned by group, because groups were recruited locally and a group mean removes the field conditions that neighbors share.

Jointly, all 25 survey and practice-session regressors explain 19 percent of the variance of the propensity of the 142 linked farmers in sample and 1 percent after adjusting for their number. In groups left out of the fit they predict none of it. Partialling the blocks out of the decision-level residuals shows where they act. Field

experience with real canal position, field water conditions, beliefs, and the practice session each account for about a tenth of the group part beyond a placebo with shuffled rows. No block accounts for the personal part beyond the placebo.

Table D6: Field experience, willingness to pay and beliefs, and buying beyond the private benchmark

	n	No controls	Within region	Within group
Field experience and real canal position				
Uses drip now	142	+0.052 (0.23)	+0.050 (0.24)	+0.009 (0.85)
Ever used drip	142	+0.032 (0.41)	+0.020 (0.63)	−0.002 (0.97)
Own plot at the head of a canal	142	−0.023 (0.64)	−0.045 (0.34)	−0.044 (0.39)
Own plot at the tail of a canal	142	+0.019 (0.68)	+0.025 (0.58)	+0.039 (0.54)
Own plot not on a canal	142	−0.003 (0.95)	−0.001 (0.99)	+0.025 (0.61)
Field water conditions				
Medium or high salinity	142	+0.013 (0.71)	−0.001 (0.97)	−0.011 (0.80)
Poor water in peak season	141	−0.003 (0.93)	+0.032 (0.42)	+0.024 (0.68)
Irrigation network in bad condition	142	−0.031 (0.36)	−0.006 (0.89)	−0.011 (0.83)
Willingness to pay				
Subsidy needed to adopt (switch point)	142	−0.001 (0.99)	−0.022 (0.59)	−0.060 (0.27)
Refuses drip at every subsidy	142	+0.037 (0.43)	+0.003 (0.95)	−0.036 (0.56)
Beliefs and attitudes				
Plans to adopt drip	131	−0.007 (0.86)	+0.026 (0.46)	+0.087 (0.09)
Sees benefits in water-saving technology	126	−0.010 (0.77)	−0.007 (0.85)	−0.004 (0.94)
Expected change in own water use with drip	101	−0.016 (0.70)	−0.041 (0.34)	−0.016 (0.77)
Expected change in yield with drip	89	−0.053 (0.15)	−0.019 (0.68)	−0.058 (0.33)
Perceived drip use in the district	142	+0.039 (0.31)	+0.036 (0.39)	+0.024 (0.60)
Expected own water use if the district adopts	128	+0.013 (0.71)	+0.011 (0.78)	+0.018 (0.71)
Confidence in these answers	142	−0.031 (0.28)	−0.048 (0.14)	−0.027 (0.46)
Risk aversion	142	+0.049 (0.13)	+0.049 (0.12)	+0.050 (0.18)
Innovativeness	142	−0.025 (0.50)	−0.022 (0.51)	−0.006 (0.88)
Social proof	142	−0.011 (0.74)	−0.012 (0.71)	−0.003 (0.92)
Farm, farmer and practice session				
Log land	142	+0.059 (0.19)	+0.054 (0.22)	+0.049 (0.47)
Agricultural cluster member	142	+0.054 (0.18)	+0.068 (0.13)	+0.061 (0.31)
Woman	142	−0.083 (0.00)	−0.075 (0.00)	−0.079 (0.02)
Fellow players known	190	−0.027 (0.39)	−0.024 (0.42)	+0.020 (0.73)
Practice-session sharing	190	+0.049 (0.10)	+0.030 (0.31)	+0.046 (0.28)
Sat at the tail in the practice session	190	+0.043 (0.13)	+0.043 (0.14)	+0.043 (0.18)
Practice-session investment	190	+0.016 (0.67)	+0.007 (0.83)	+0.112 (0.06)

Notes: Coefficients per standard deviation of the characteristic, with p-values from game-clustered standard errors in parentheses. The outcome is the mean over the two first decisions of the purchase minus the private benchmark (standard deviation 0.42). No farmer is dropped for a question that did not apply. Canal position has four categories for all 142 linked farmers: head, middle (the reference), tail, and a main plot not on a canal, for which the question was not asked; each row is estimated with the other two categories. Sharing is undefined for the 38 farmers who sat at the tail in the practice session, since no one was below them. They keep the mean value, and the indicator is estimated jointly with sharing. Survey measures come from the main survey; most linked interviews preceded the games, but not all (Appendix A). The switch point is the subsidy step at which a farmer would adopt, with refusals at every step set to 25; higher means a larger subsidy is needed. Expected changes with drip were asked only of farmers who had never used drip and grew certain main crops. All estimates are associations.

Table D7 relates choosing alike to the social ties among the players, using raw first-decision choices without the benchmark or fixed effects.

The decomposition of sharing in Table 5 uses the same moments. Its unit is the farmer-session mean of the residual excess over a fair share of the water left, at seats 1–4. The residual removes the mean at the same session, seat, card and water band

Table D7: Social ties and choosing alike

A. Groups, first decisions of Sessions 2 and 3	Groups	Unanimous	Pairs choosing alike	Buying rate
Mostly strangers (fewer than 1.5 known)	15	0.13	0.50	0.50
Some ties (1.5 to 3 known)	11	0.55	0.75	0.39
Close-knit (more than 3 known)	12	0.25	0.61	0.43
Three or more members name a relative, neighbor or friend	10	0.40	0.67	0.39
Fewer close ties	28	0.25	0.58	0.46

B. Farmers, when most of the other four made one choice	Decisions	Share matching the majority
Knows none of the other four	48	0.50
Knows 1 to 3	111	0.62
Knows all four	95	0.68
Names a relative, neighbor or friend among them	83	0.77
No such tie	171	0.55
Independent choices at the observed buying rates		0.51

Notes: Ties come from the questionnaire after the games: how many of the other four players a farmer knew, and the relationship to each. Group categories use the group mean of players known; we chose the cut-offs after seeing the data. Unanimous: all five made the same first-decision choice. Pairs choosing alike: share of the ten pairs of members with the same choice. Panel B excludes decisions in which the other four split two against two. Permutation p-values shuffle tie categories across groups (panel A) or farmers (panel B), 5,000 draws: some ties against mostly strangers, unanimity $p = 0.005$ and pairs $p = 0.003$; any ties against mostly strangers, $p = 0.04$ and $p = 0.02$; knows all four against none, $p = 0.05$; close tie against none, $p = 0.002$. In a separate test, district field conditions (17 districts, from the 914-farm survey), the composition of the group and its play in the practice session predict the group part no better than chance: 5 of 132 tests have $p < 0.05$, and no set forecasts groups left out of the fit. All estimates are associations.

(up to 6.5, 6.5–9.5, 9.5–13.5 and above 13.5 units). Of 186 farmers with a sharing measure, 118 have one in both paid sessions; the others sat at the tail in one session. The personal part has a permutation $p < 0.001$ when Session 3 measures are shuffled within groups. The tie result regresses a farmer's residual on the mean residual of the other members in the same session, separately by ties, and shuffles ties across farmers (2,000 draws).

Table D8 relates sharing to the same characteristics as buying.

Maintenance behaves differently from both. Farmers contributed 14.4 tokens per round on average, close to the 15 tokens the rules describe as maintaining supply, at every seat. In most of Uzbekistan, farmers clean and repair their canals together through hashar, unpaid communal work, in time for the spring planting season [1, 10]. Cooperative maintenance in the game is therefore not surprising. The same decomposition gives seat and session almost none of the variance of a farmer's mean contribution in a session (0.01). Of the part that follows a farmer, a group or a region, 55 percent is personal, 35 percent belongs to the group, much of it persistent across sessions, and 10 percent to the region.

Table D8: Who takes more than a fair share

	n	No controls	Within region	Within group
Field experience and real canal position				
Uses drip now	138	+0.073 (0.25)	+0.055 (0.37)	+0.042 (0.58)
Ever used drip	138	+0.058 (0.29)	+0.081 (0.12)	+0.043 (0.53)
Own plot at the head of a canal	138	−0.110 (0.05)	−0.073 (0.18)	−0.082 (0.30)
Own plot at the tail of a canal	138	−0.024 (0.75)	−0.034 (0.64)	−0.079 (0.41)
Own plot not on a canal	138	+0.105 (0.18)	+0.070 (0.35)	+0.052 (0.43)
Field water conditions				
Medium or high salinity	138	−0.043 (0.49)	+0.019 (0.76)	−0.008 (0.92)
Poor water in peak season	137	+0.115 (0.06)	+0.068 (0.34)	+0.114 (0.12)
Irrigation network in bad condition	138	−0.002 (0.98)	−0.073 (0.29)	−0.045 (0.61)
Willingness to pay				
Subsidy needed to adopt (switch point)	138	+0.045 (0.52)	+0.134 (0.04)	+0.098 (0.22)
Refuses drip at every subsidy	138	−0.075 (0.31)	+0.043 (0.65)	+0.083 (0.44)
Beliefs and attitudes				
Plans to adopt drip	127	+0.104 (0.08)	+0.014 (0.82)	+0.110 (0.24)
Sees benefits in water-saving technology	122	−0.075 (0.29)	−0.084 (0.22)	−0.026 (0.72)
Expected change in own water use with drip	99	−0.147 (0.01)	−0.111 (0.18)	−0.043 (0.67)
Expected change in yield with drip	87	+0.017 (0.86)	−0.073 (0.31)	−0.082 (0.31)
Perceived drip use in the district	138	+0.049 (0.47)	+0.069 (0.31)	+0.057 (0.44)
Expected own water use if the district adopts	124	−0.145 (0.02)	−0.140 (0.01)	−0.057 (0.44)
Confidence in these answers	138	−0.062 (0.23)	−0.044 (0.50)	−0.052 (0.50)
Risk aversion	138	−0.089 (0.26)	−0.082 (0.23)	−0.027 (0.67)
Innovativeness	138	−0.066 (0.37)	−0.086 (0.25)	−0.081 (0.30)
Social proof	138	−0.068 (0.18)	−0.089 (0.04)	−0.021 (0.69)
Farm, farmer and practice session				
Log land	138	−0.083 (0.29)	−0.067 (0.34)	−0.063 (0.50)
Agricultural cluster member	138	−0.049 (0.34)	−0.034 (0.44)	+0.010 (0.86)
Woman	138	+0.118 (0.04)	+0.075 (0.22)	+0.007 (0.93)
Fellow players known	186	−0.117 (0.10)	−0.085 (0.18)	−0.029 (0.70)
Practice-session sharing	186	−0.294 (0.00)	−0.229 (0.00)	−0.259 (0.00)
Sat at the tail in the practice session	186	−0.031 (0.55)	−0.032 (0.53)	−0.029 (0.61)
Practice-session investment	186	−0.026 (0.77)	+0.032 (0.70)	+0.155 (0.13)

Notes: The outcome is a farmer's mean excess over a fair share of the water left, at seats 1–4 of the paid sessions, after removing the mean at the same session, seat, card and water band (standard deviation 0.80 units). Positive values mean taking more, that is, sharing less. Coefficients per standard deviation of the characteristic, with p-values from game-clustered standard errors in parentheses. Every farmer is kept: canal position has four categories (middle is the reference) and farmers who sat at the tail in practice keep an indicator, as in Table D6. All estimates are associations.

Table D9 relates maintenance to the same characteristics. Contributions in the practice session predict contributions in the paid sessions ($p < 0.01$ in every specification). Women and farmers who have used drip contribute more, and more risk-averse farmers less. Within groups, farmers who refuse drip at every subsidy contribute less, by 0.6 tokens per standard deviation ($p = 0.01$).

Table D9: Who contributes more to maintenance

	n	No controls	Within region	Within group
Field experience and real canal position				
Uses drip now	142	+0.316 (0.24)	+0.317 (0.10)	+0.201 (0.30)
Ever used drip	142	+0.437 (0.05)	+0.311 (0.06)	+0.188 (0.34)
Own plot at the head of a canal	142	−0.046 (0.86)	−0.296 (0.21)	−0.168 (0.63)
Own plot at the tail of a canal	142	−0.179 (0.57)	−0.063 (0.83)	−0.112 (0.75)
Own plot not on a canal	142	+0.068 (0.86)	+0.163 (0.66)	−0.329 (0.48)
Field water conditions				
Medium or high salinity	142	+0.174 (0.54)	−0.051 (0.86)	+0.180 (0.53)
Poor water in peak season	141	−0.399 (0.15)	−0.165 (0.61)	−0.054 (0.91)
Irrigation network in bad condition	142	−0.159 (0.54)	+0.148 (0.66)	+0.252 (0.50)
Willingness to pay				
Subsidy needed to adopt (switch point)	142	+0.181 (0.44)	−0.019 (0.94)	−0.124 (0.68)
Refuses drip at every subsidy	142	+0.013 (0.96)	−0.636 (0.05)	−0.592 (0.01)
Beliefs and attitudes				
Plans to adopt drip	131	−0.323 (0.28)	+0.028 (0.94)	+0.073 (0.88)
Sees benefits in water-saving technology	126	−0.166 (0.42)	−0.080 (0.70)	−0.101 (0.72)
Expected change in own water use with drip	101	+0.700 (0.04)	+0.467 (0.13)	+0.440 (0.23)
Expected change in yield with drip	89	−0.048 (0.82)	+0.327 (0.20)	+0.343 (0.35)
Perceived drip use in the district	142	+0.327 (0.22)	+0.307 (0.21)	+0.256 (0.31)
Expected own water use if the district adopts	128	−0.019 (0.94)	−0.114 (0.68)	−0.108 (0.74)
Confidence in these answers	142	+0.423 (0.22)	+0.301 (0.39)	−0.002 (1.00)
Risk aversion	142	−0.479 (0.03)	−0.486 (0.04)	−0.167 (0.50)
Innovativeness	142	−0.026 (0.90)	+0.024 (0.91)	−0.133 (0.60)
Social proof	142	−0.051 (0.78)	−0.035 (0.83)	−0.177 (0.34)
Farm, farmer and practice session				
Log land	142	+0.191 (0.42)	+0.129 (0.60)	+0.177 (0.57)
Agricultural cluster member	142	−0.141 (0.60)	−0.075 (0.75)	−0.097 (0.74)
Woman	142	+0.449 (0.01)	+0.617 (0.00)	+0.241 (0.10)
Fellow players known	190	−0.208 (0.44)	−0.246 (0.34)	+0.527 (0.24)
Practice-session sharing	190	+0.097 (0.72)	−0.368 (0.12)	+0.067 (0.82)
Sat at the tail in the practice session	190	+0.186 (0.35)	+0.186 (0.37)	+0.186 (0.41)
Practice-session investment	190	+1.126 (0.00)	+1.011 (0.00)	+1.786 (0.00)

Notes: The outcome is a farmer's mean maintenance contribution per round in the paid sessions, after removing the mean at the same session and seat (standard deviation 3.1 tokens). Coefficients per standard deviation of the characteristic, with p-values from game-clustered standard errors in parentheses. Every farmer is kept: canal position has four categories (middle is the reference) and farmers who sat at the tail in practice keep an indicator, as in Table D6. All estimates are associations.

E Survey link and field context

The exact game-node crosswalk links 142 of 190 participants to 139 survey households; 48 links remain unknown. The linked sample includes three pairs from a common household, each counted as participants. The 914-farm comparison counts each survey household once. No new identity matching is performed, and unknown links are not coded as nonusers.

Table E1: Linked participants beside the full farm survey

Region	Sample	n	Land (ha)	Women (%)	Salinity (%)	Ever used drip (%)
All	Linked	142	33.5	7.7	42.3	12.0
All	Survey	914	32.0	8.0	48.6	11.4
Fergana	Linked	46	26.9	13.0	37.0	2.2
Fergana	Survey	232	34.4	10.8	53.9	4.3
Kashkadarya	Linked	24	30.0	4.2	37.5	8.3
Kashkadarya	Survey	227	35.0	14.5	44.5	7.0
Samarkand	Linked	33	42.0	12.1	27.3	18.2
Samarkand	Survey	229	22.0	5.2	27.5	14.4
Khorazm	Linked	39	46.0	0.0	64.1	20.5
Khorazm	Survey	226	42.0	1.3	68.6	19.9

Notes: Every displayed characteristic has the row's explicit denominator n; no item is missing in these selected fields. Land is median total farm area; salinity means medium or high on plot 1. Ever used drip is the impact evaluation's adoption measure (module D2), the same definition as the field rates in Table 10. An earlier measure from crop irrigation-method records disagrees with it for 68 of the 914 farms and is not used. The linked participants comprise 46/50 in Fergana, 24/55 in Kashkadarya, 33/40 in Samarkand and 39/45 in Khorazm; the 48 unknown links are excluded from characteristics. Only non-identifying workbook fields were read; keys remained in memory and only aggregates were exported.

The survey chronology is documented in Appendix A; the linked interviews are not uniformly pregame. The sample's regional coverage and farm characteristics limit individual extrapolation. The comparison of real drip users in Section 7.3 is an association within the linked sample, based on 17 farmers who have used drip, 11 of them currently. Agricultural cluster membership is an institutional arrangement, not a social-network measure.

E.1 Purchases beyond the incentives of position

For each farmer we sum, over the two first decisions, the purchase minus the rate predicted by randomized position alone (seat-by-session means). This is purchases beyond the incentives of position, with a standard deviation of 0.84. About a third of its variance (0.34) lies between groups, against about 0.20 when farmers are randomly regrouped within region. Table D6 relates the same propensity, measured against

the private benchmark, to field experience, real canal position, field conditions, willingness to pay, beliefs and the practice session. It uses three levels of control and keeps every farmer. Section 5 divides the same residual into personal and group parts (Appendix D.5).

With 142 linked farmers, few associations are individually precise. The signs that agree across all three specifications fit the canal story. Current drip users, farmers at the tail of a real canal, larger farms, cluster members, more risk-averse farmers and farmers who shared more in practice buy more. Farmers at the head and the 11 women buy less. Salinity, seasonal water, network condition, willingness to pay and beliefs about drip show no consistent association.

For the outcomes of Table 9, the other characteristics differ as follows:

- cluster members buy drip 13–17 percentage points more often at the first decision (p between 0.06 and 0.11) and do not differ in water use or maintenance;
- plans to adopt drip and refusal of drip at every subsidy are associated with practice-session extraction only across regions;
- within groups, farmers who refuse drip at every subsidy invest about 1.3 tokens less per round in the paid sessions.

These results are exploratory.